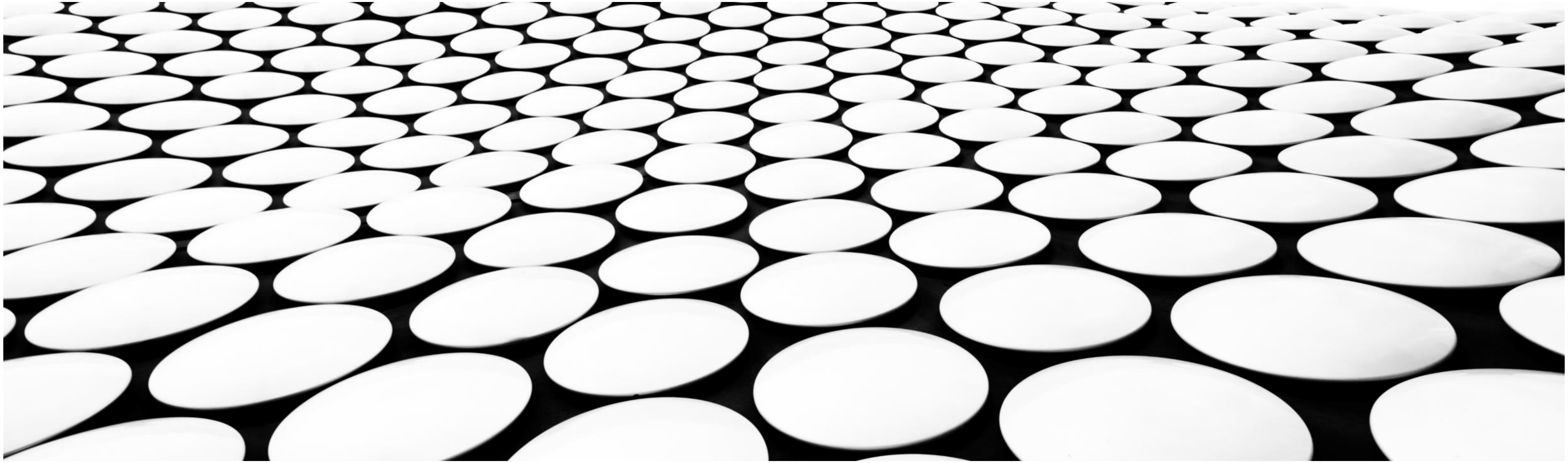
EXAMINING NETWORK LAYERS 1 & 2

WEEK 02



AGENDA

At the end of this week the learners should be able to:

- Explain the services provided by Layers 1 and 2
- Describe the TCP/IP protocols commonly used at Network Access Layer specifically: ARP, Ethernet (IEEE802.3), WLAN and WiFi (IEEE802.11), Bluetooth (IEEE802.15) and WiMax (IEEE802.16)
- Simulate how ARP works in physical addressing and examine Ethernet frames
- Simulate how switches work to facilitate network connectivity using Packet Tracer

UNDERSTANDING NETWORK ACCESS (LAYER 1 & 2)

- ❑ Media
- ❑ Topology
- ❑ Media Access Control
- ❑ Address Resolution Protocol (ARP)

MEDIA

LINKS

- The transmission media (often referred to in the literature as the physical media) used to link devices to form a computer network.
- The main network media categories are:
  - Wired (guided) e.g. coaxial cable, twisted pair, optical fiber
  - Wireless (unguided) e.g. WiFi, Bluetooth, WiMAX, microwave, satellite

TWISTED PAIR CABLE

- Four pairs of copper wires twisted around each other
- Created in 1881 by Alexander Graham Bell
- Telephones wires: two wires, one for sending the signal and one for receiving it
- They twisted the wires at each junction to resist the electro-magnetic interference. This is because telephone wires shared same route as power lines.
- Tighter twisting gives higher quality wires
- Uses RJ45 connectors
- TIA/EIA-568 contains cabling specifications

TWISTED PAIR CABLE

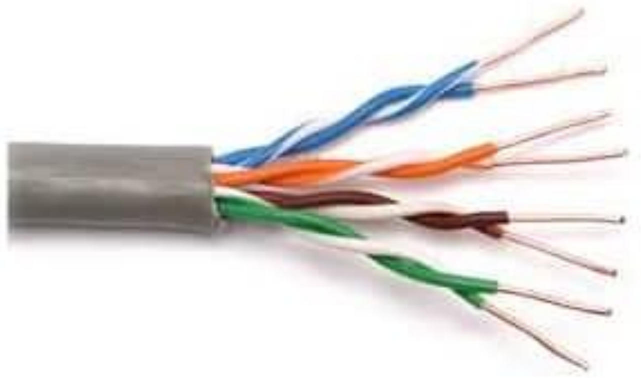
Two types

Shielded (STP)

- It has an insulating material around each pair and another overall metallic foil/braid that helps further resist the interference.
- Reduces electrical noise within cable (pair-to-pair coupling, or crosstalk) and from outside the cable (EMI and RFI).
- More expensive and difficult to install than UTP
- Shielded is less common,

Unshielded (UTP)

- Unshielded is the dominant cable



UTP Cable

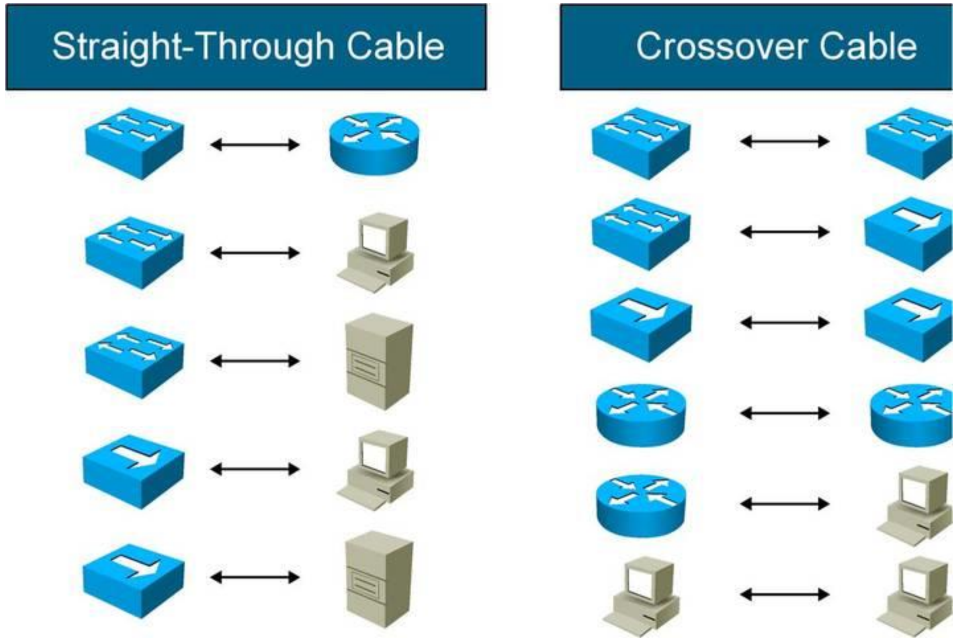
STP Cable

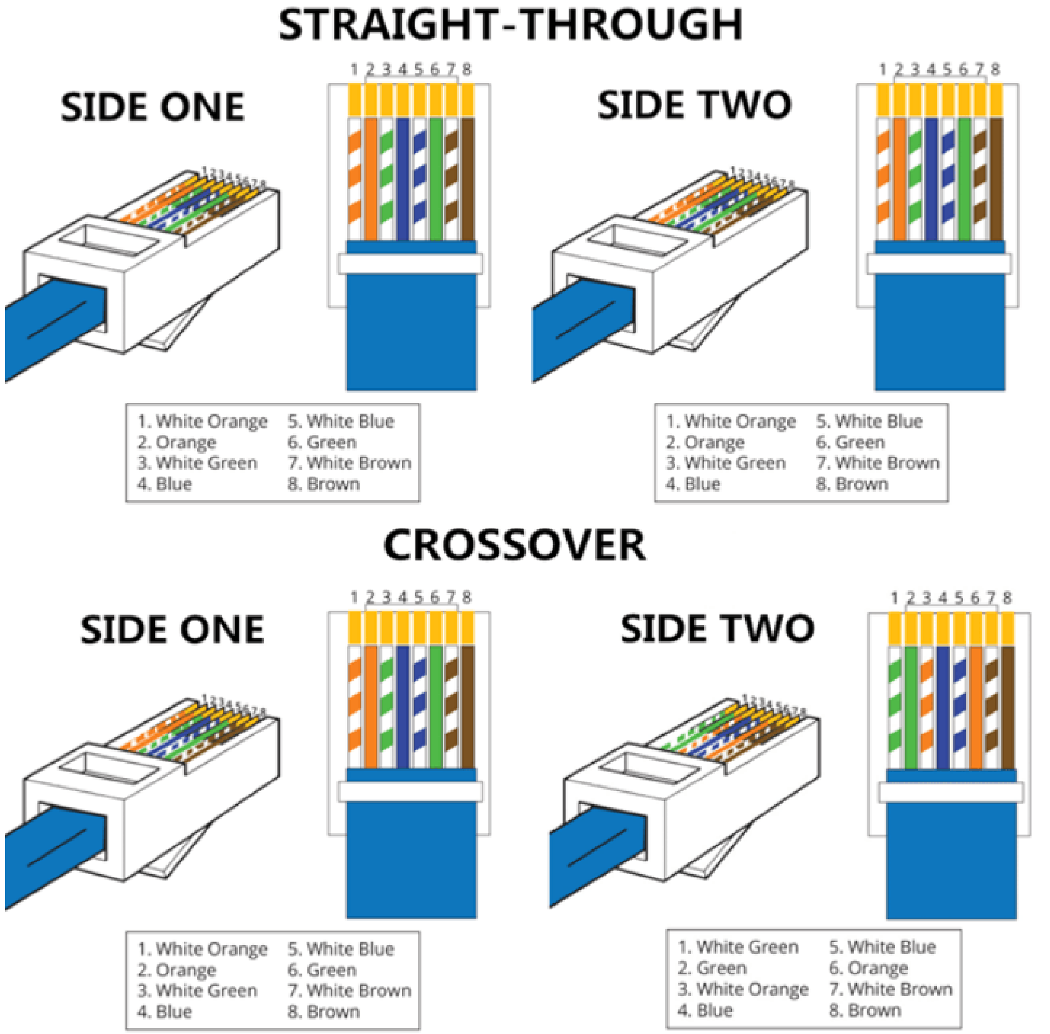
TWISTED PAIR CABLE

Categories (Cat)

Category	Speed	Frequency
CAT 1	Carry only voice	1MHz
CAT 2	4Mbps	4MHz
CAT 3	10Mbps	16Mhz
CAT 4	16Mbps	20Mhz
CAT 5	100Mbps	100Mhz
CAT 5e	1000Mbps/1Gbps	100Mhz
CAT 6	1000Mbps/1Gbps	250MHz
CAT 7	10Gbps	600MHz
CAT 7a	10Gbps	1000Gbps
CAT 8	25Gbps	2000Mhz

TWISTED PAIR CABLE



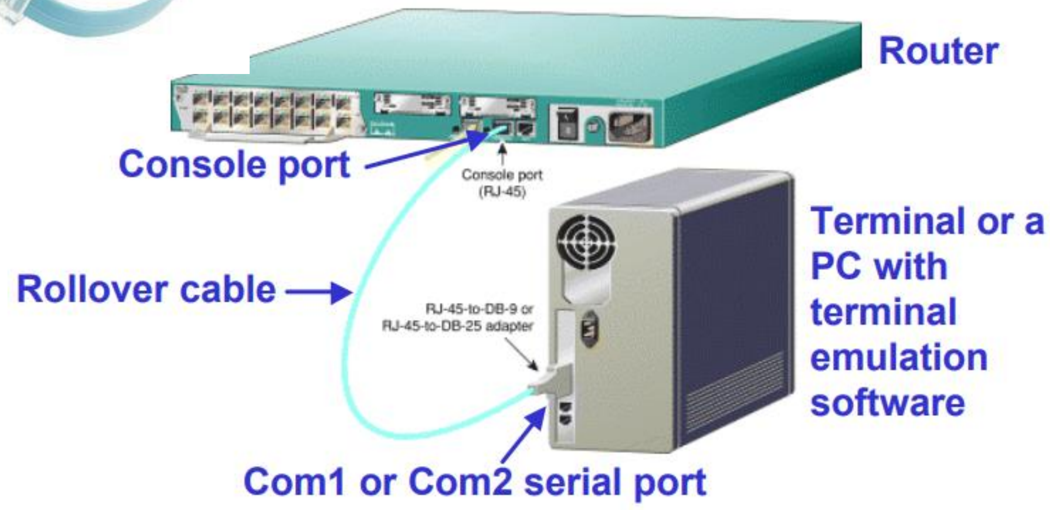


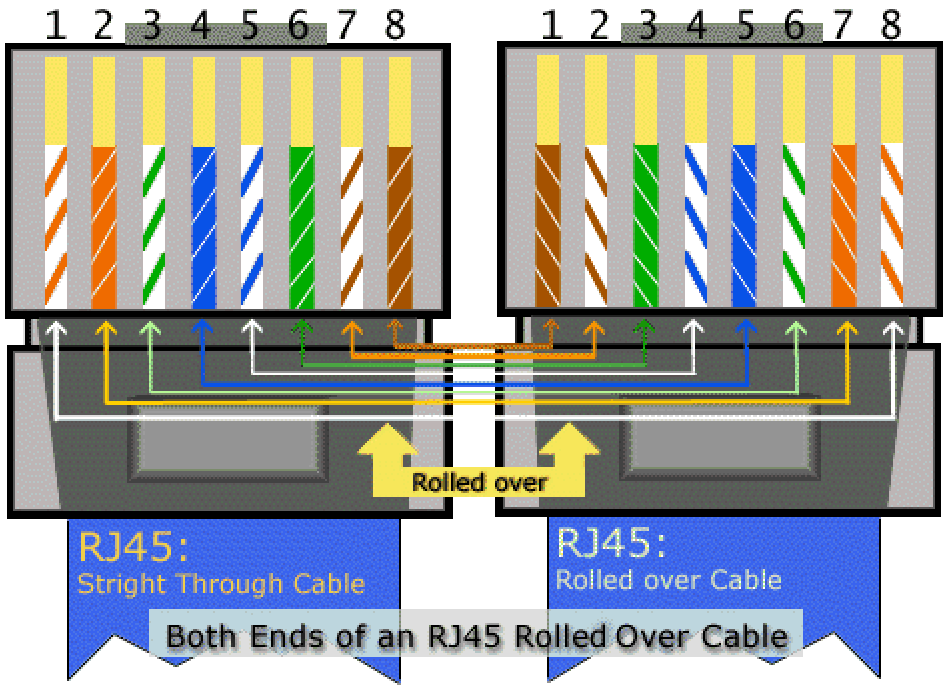
Straight-through aka Patch Cable - Connects computer NIC port to the switch port

Crossover Cable - Connects switch port to another switch port

TWISTED PAIR CABLE

- Rollover Cable connects the RJ-45 adapter on the com port of the computer to the console port of the router or switch





TWISTED PAIR CABLE

Advantages

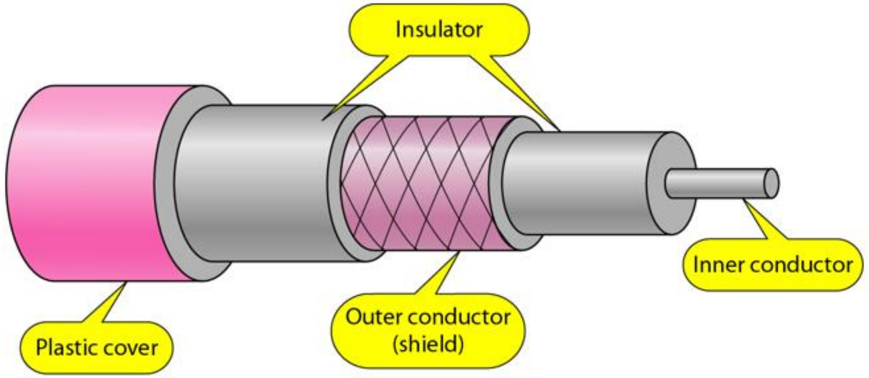
- Installation is easy
- Flexible
- Cheap
- It has high speed capacity
- Higher grades of UTP are used in LAN technologies like Ethernet.

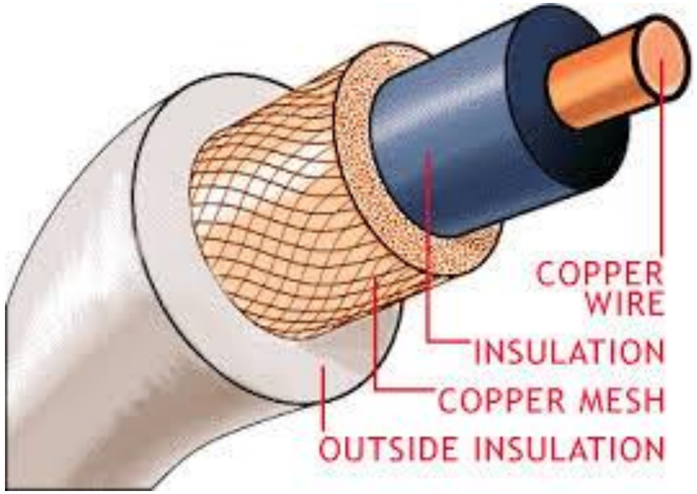
Disadvantages

- 100 meter limit
- Bandwidth is low when compared with Coaxial Cable and Fiber
- Provides less protection from interference.

COAXIAL (COAX) CABLE

- Invented by Oliver Heaviside in 1880. Co: Stands for together, coordination. Axial: stands for axis i.e. all centered around the same point
- Consists of single inner wire located in the center made of copper, surrounded by a layer of flexible insulation. Over this insulating material is a woven copper braid or metallic foil that acts both as the second wire in the circuit and as a shield for the inner conductor. This second layer (shield) helps reduce the amount of outside interference. An outer jacket covers this shield.

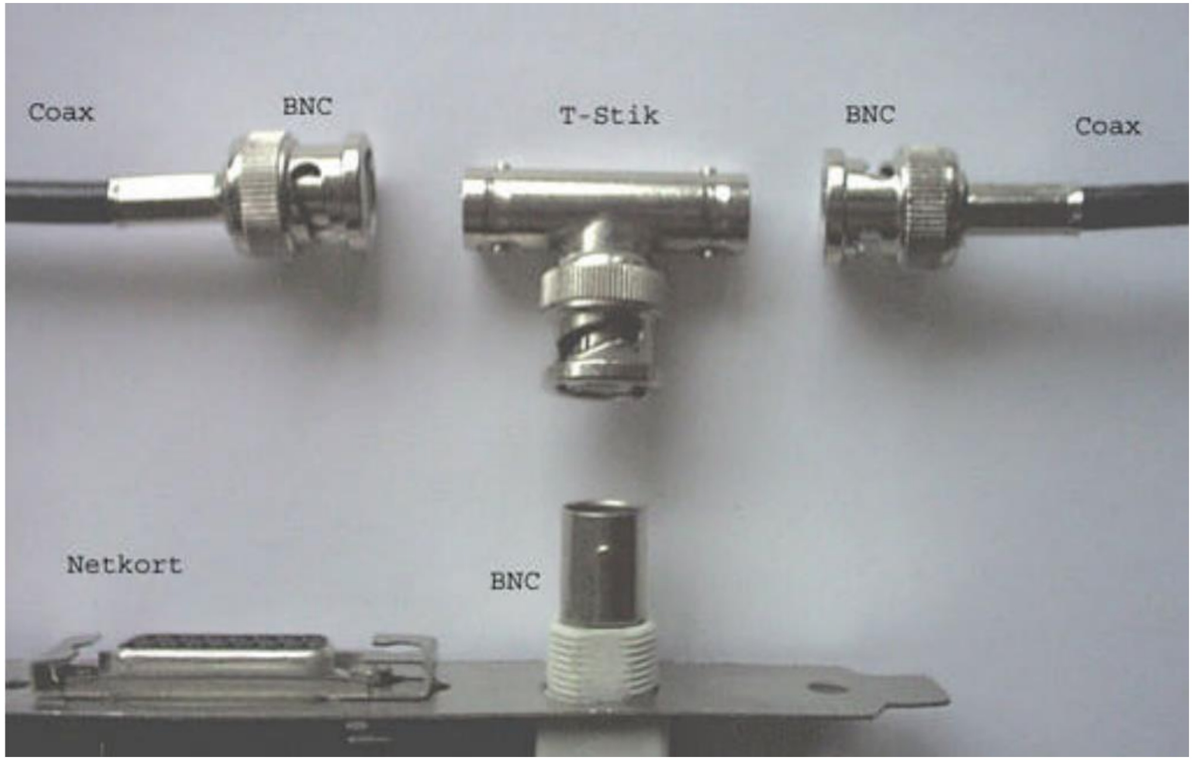




COAXIAL (COAX) CABLE

- The BNC connector shown looks like a cable-television connector and connects to an older NIC with a BNC interface.
- Categories

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet



COAXIAL (COAX) CABLE

Advantages

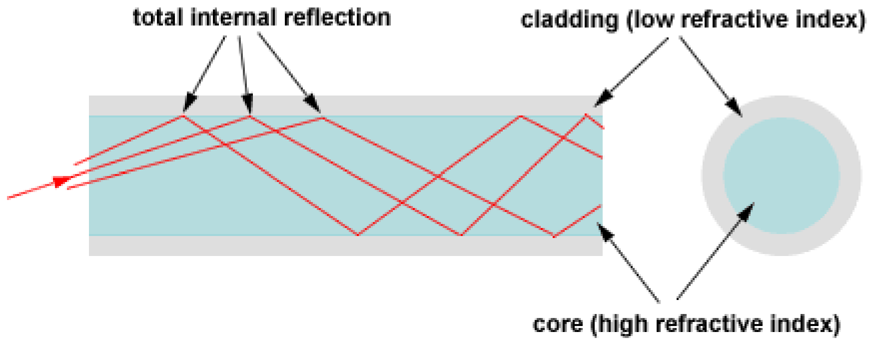
- Better shielding and more bandwidth for longer distances and higher rates than twisted pair.
- Ethernet can run approximately 100 meters using twisted-pair cable, but 500 meters using coaxial cable.

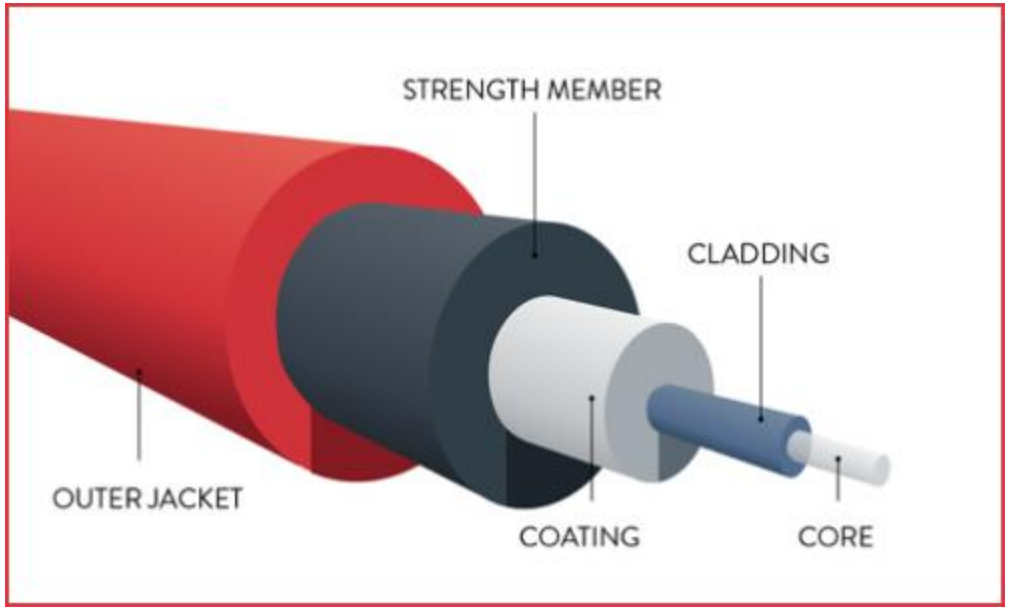
Disadvantages

- More difficult to install
- More expensive than TP.

FIBER-OPTIC CABLE

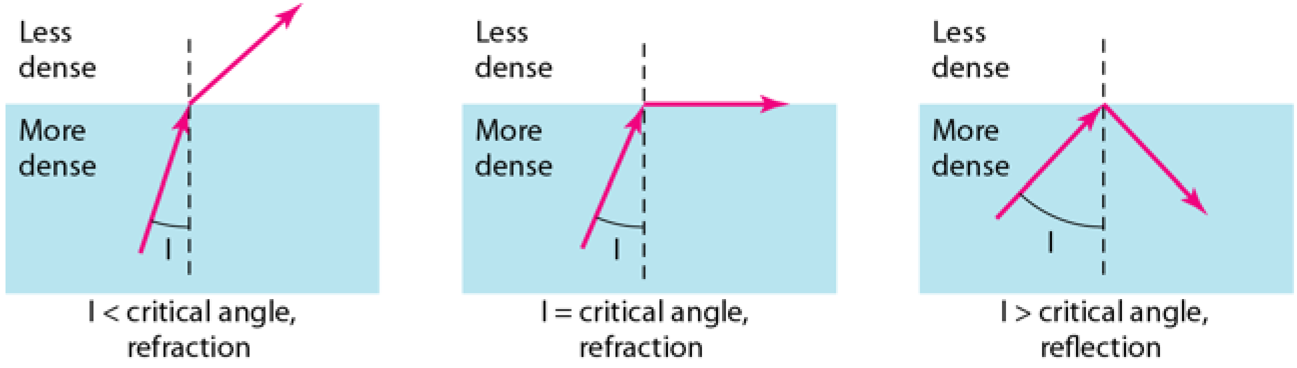
- Uses plastic or glass fibers to transmit signals in form of light at fast speeds.
- Optical fibers use reflection to guide light through a channel.
- A glass or plastic core is surrounded by a cladding of less dense glass or plastic.
- The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted. This process is called total internal reflection.





FIBER-OPTIC CABLE

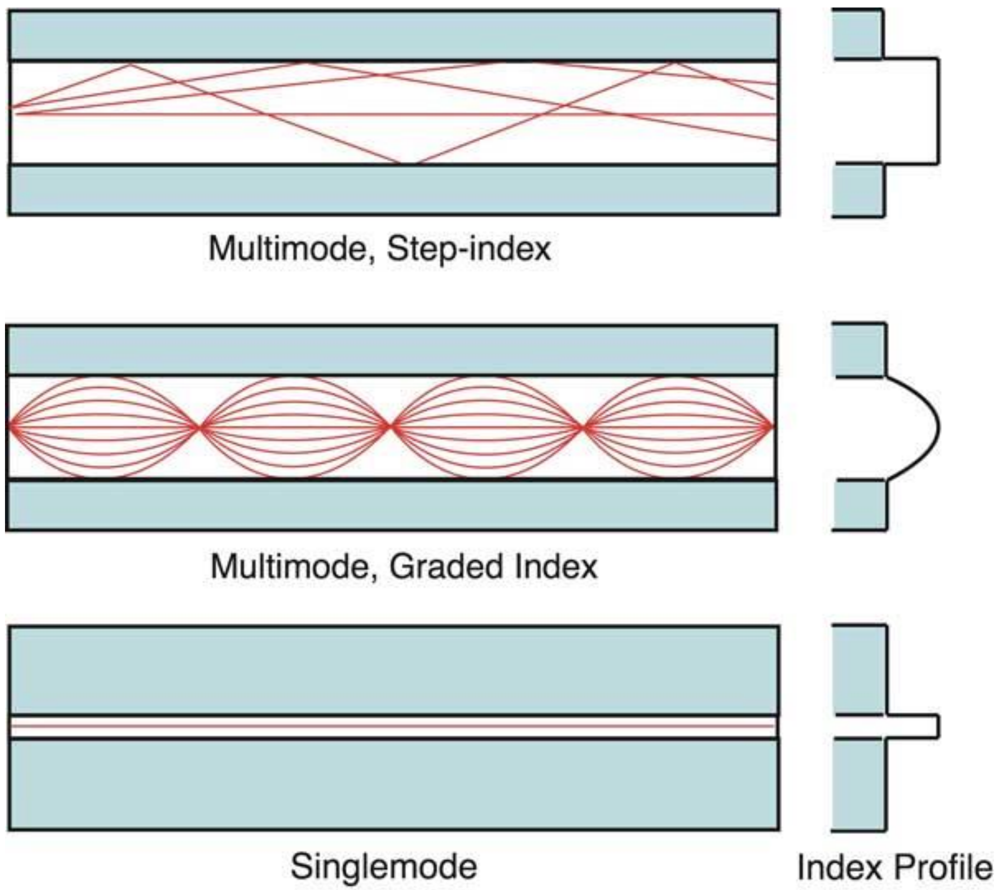
- Light travels in a straight line as long as it is moving through a single uniform substance.
- If a ray of light traveling through one substance suddenly enters another (less or more dense) substance, its speed changes abruptly, causing the ray to change direction. This change is called refraction.
- If the angle of incidence increases, so does the angle of refraction.
- The critical angle is defined to be an angle of incidence for which the angle of refraction is 90 degrees.



FIBER-OPTIC CABLE

Two types of fiber-optic cables exist:

- Single-mode: allows only one mode (or wavelength) of light to propagate through the fiber. Uses lasers as the light-generating method and is more expensive than multimode cable. The maximum cable length of single-mode cable is 60+ km (37+ miles). Gives higher bandwidth and greater distances than multimode and is often used for campus backbones.
- Multimode: allows multiple modes of light to propagate through the fiber. Uses light emitting diodes (LEDs) as light-generating devices. The maximum length of multimode cable is 2 km (1.2 miles). Often used for LAN.



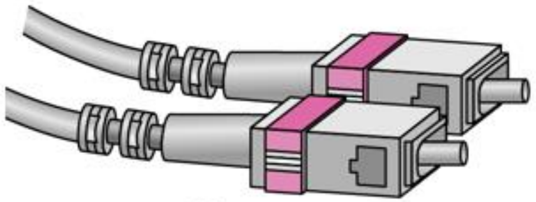
WARNING: Laser is so strong that it can damage eyes. Never look at the end of a fiber or the transmit port on a NIC, switch, or router

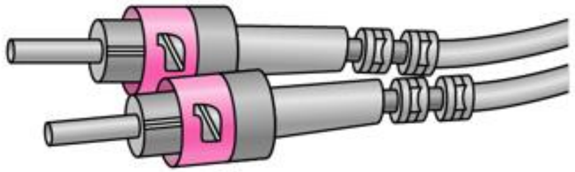
FIBER-OPTIC CABLE

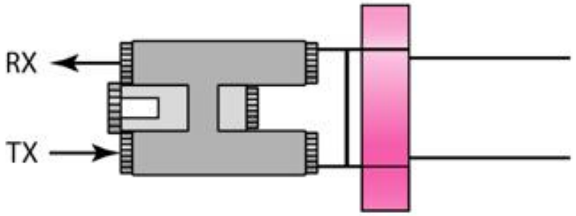
- Optical fibers are defined by the ratio of the diameter core to the diameter of their cladding, both expressed in micrometers. The common types are:

Type	*Core (µm)*	*Cladding (µm)*	*Mode*
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

- The 3 types of connectors are:


SC connector


ST connector



MT-RJ connector

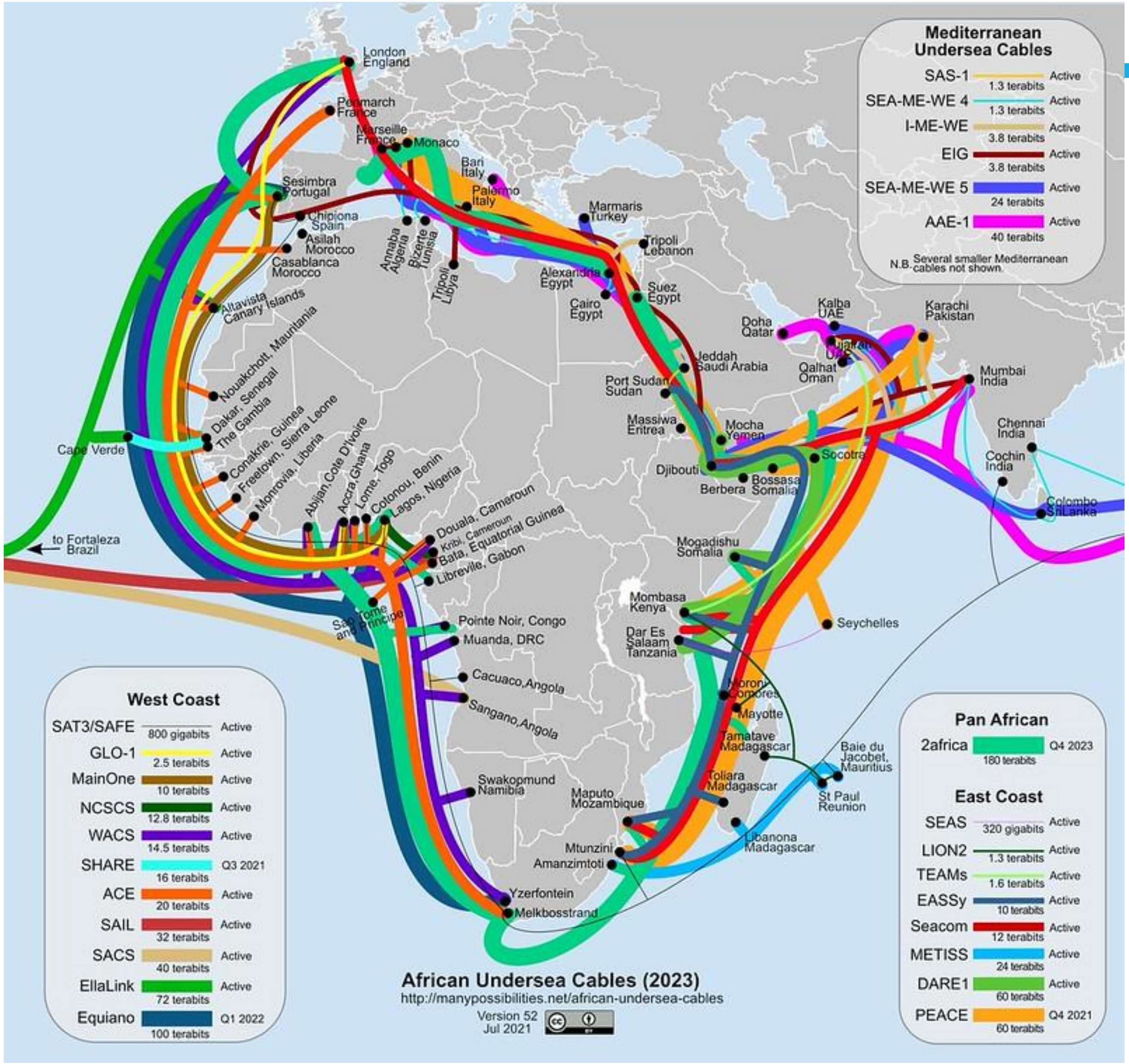
FIBER-OPTIC CABLE

Advantages

- Higher bandwidth
- Less signal attenuation
- Immunity to electromagnetic interference
- Light weight
- Greater immunity to tapping therefore more secure
- Can carry larger amounts of data over larger distances at higher speeds
- Used as the network backbone

Disadvantages

- Most expensive in cable costs, installation and maintenance



http://manypossibilities.net/african-undersea-cables

TEAMS cable

- The East Africa Marine Systems (TEAMS)
- landed Msa 12 June 2009
- initial capacity 40 Gb/s upgradeable to 640 Gbit/s

SEACOM cable

- Landed in July 2009
- capacity of 1.28 Tbit/s, of which 100 Gbit/s is currently active

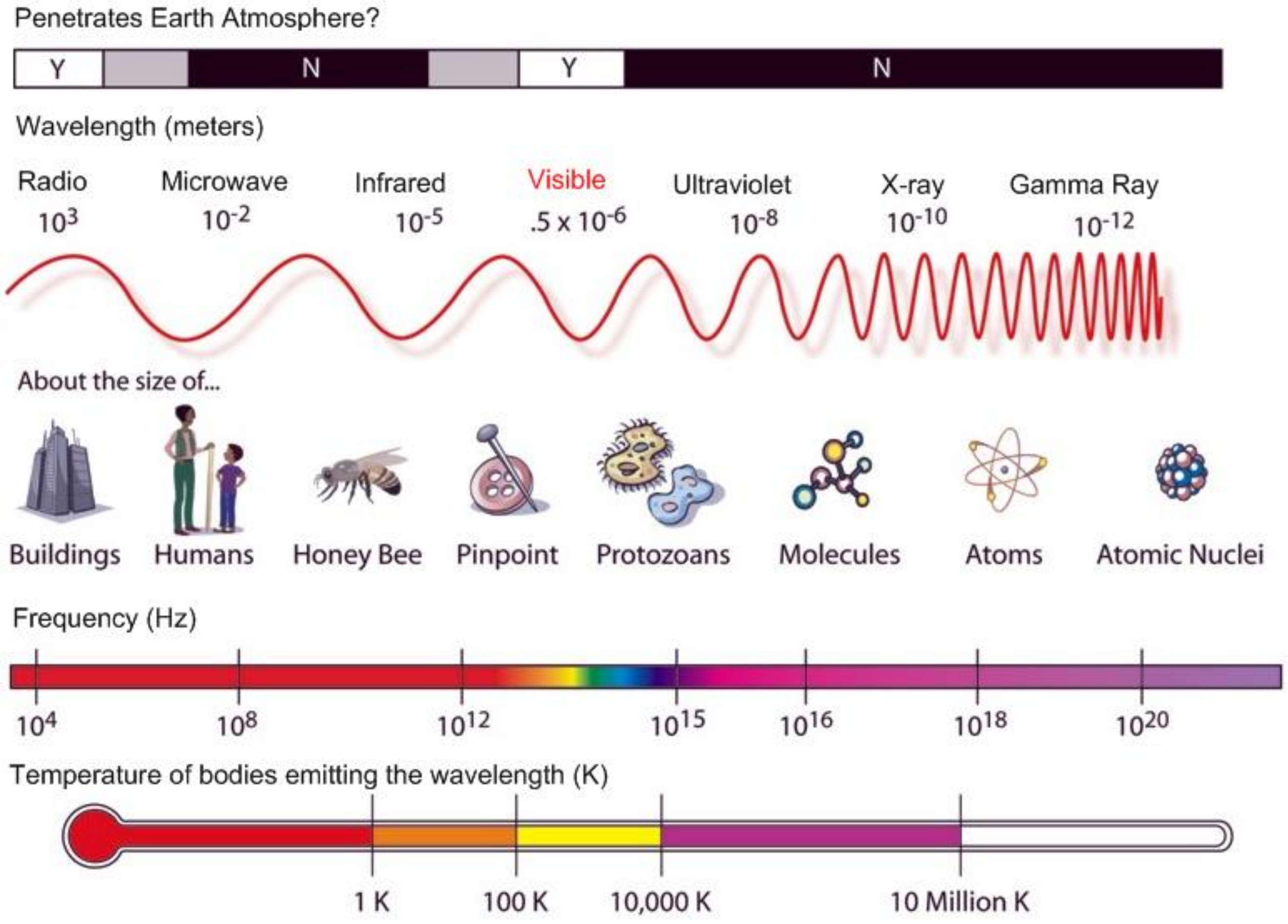
EASSy cable

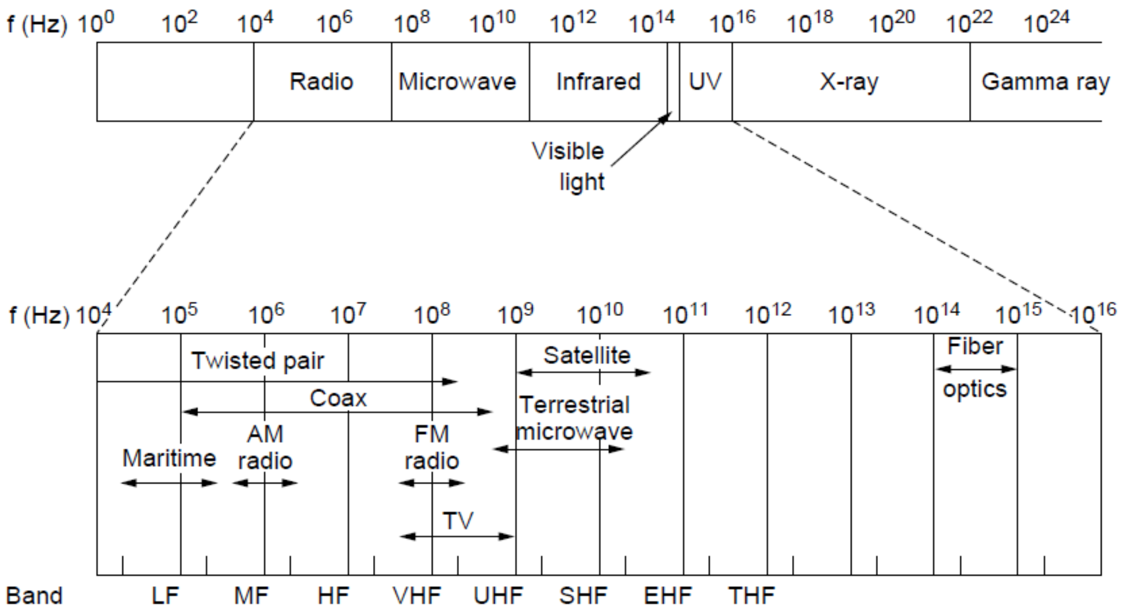
- Eastern Africa Submarine Cable System
- Landed in 16 July 2009
- 3.84 (Tbit/s)
- highest capacity system serving sub-Saharan Africa
- direct connectivity between East Africa and Europe/ North America

WIRELESS

- Wireless (unbounded) media is a type of media that is used to transmit data from one point to another without using physical wire connections.
- Use different frequencies of the electromagnetic spectrum
- Travels at the speed of light.
- A transmitting antenna and a receiver aerial used to facilitate communication.
- Examples of wireless communication media include:
  - Radio waves (low frequency, broadcast over wide area, long distances)
  - Microwaves (used for LAN and cellular communications)
  - Infrared (line of sight)
- WLANs typically use radio waves (e.g. 902 MHz), microwaves (e.g. 2.4 GHz), and infrared (IR) waves (e.g. 820 nm). We will cover this in a separate week.

The Electromagnetic Spectrum





WIRELESS

Advantages

- Does not require installation of media. Makes it easy and quick to setup network, looks good without cables and often cheaper to connect multiple users.
- Mobility: users can move around freely within the area of the network with their laptops, handheld devices
- Scalable: easy for users to join network

Disadvantages

- Affected by propagation environment effects e.g. reflection, obstruction by objects, interference.
- Lower data transfer speeds as compared to wired networks
- Less geographical distance coverage
- Less secure, hackers can intercept and sniff data as long as signal is within reach

SUMMARY

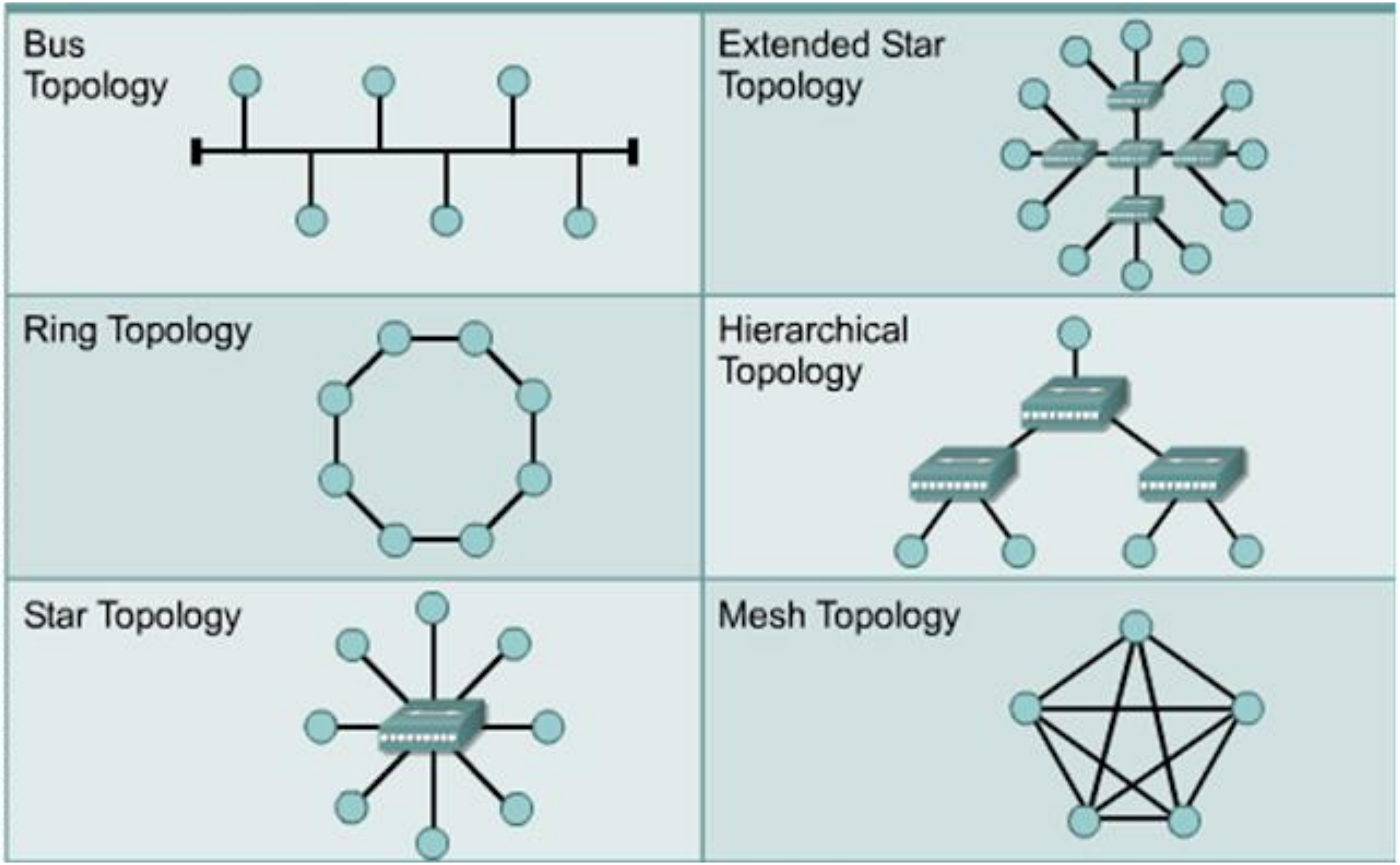
	Twisted Pair	Coaxial	Fiber Optic	Wireless LAN
Bandwidth	Up to 1 Gbps	10–100 Mbps	Up to 10 Gbps or higher	Up to 54 Mbps
Distance	Up to 100 m	Up to 500 m	Up to 60 km	Up to 100 m
Price	Least expensive	Inexpensive	Most expensive	Moderate

TOPOLOGY

TOPOLOGY

- Topology is derived from two Greek words topo and logy, where topo means 'place' and logy means 'study
- Network topology is the arrangement (also layout, structure) of the elements of a communication network.
- Physical topology is the placement of the various components of a network (e.g., device location and cable installation), while logical topology illustrates how data flows within a network.
- A network topology diagram helps visualize the communicating devices, which are modeled as nodes, and the connections between the devices, which are modeled as links between the nodes. This is an application of graph theory.

TOPOLOGY


Bus Topology
Extended Star Topology
Ring Topology
Hierarchical Topology
Star Topology
Mesh Topology

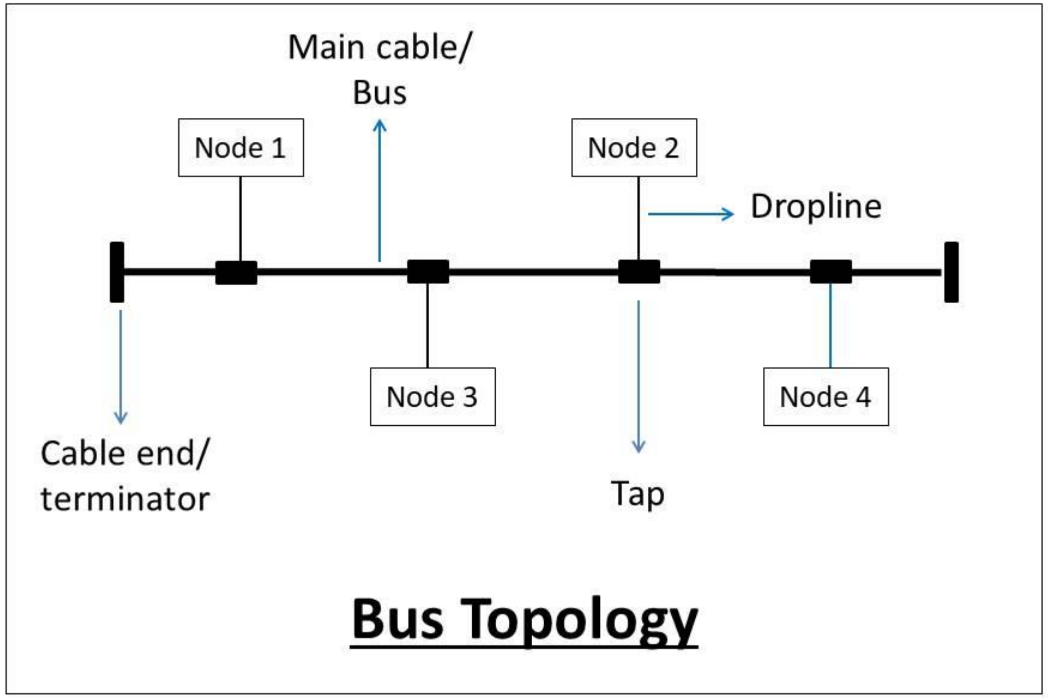
TOPOLOGY

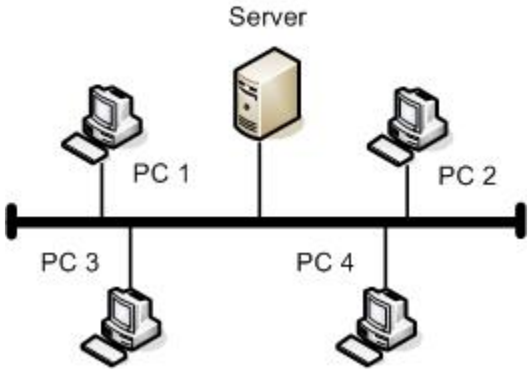
Need to be consider the following when selecting a physical topology:

- Ease of Installation
- Cabling Required
- Implementation Cost
- Maintenance Required
- Ease of Reconfiguration and upgradation
- Reliable Nature
- Fault Tolerance

BUS TOPOLOGY

- All the network nodes are connected to a common network media called the bus. The main cable acts as a spine for the entire network.
- Taps are the connectors, while droplines are the cables connecting the bus with the computer
- Only one node can receive and transmit data at a time.
- Any device that wants to communicate with other device on the network will send its data over the bus which will be send to all attached devices but only the intended recipient will only process that packet.





BUS TOPOLOGY

Advantages:

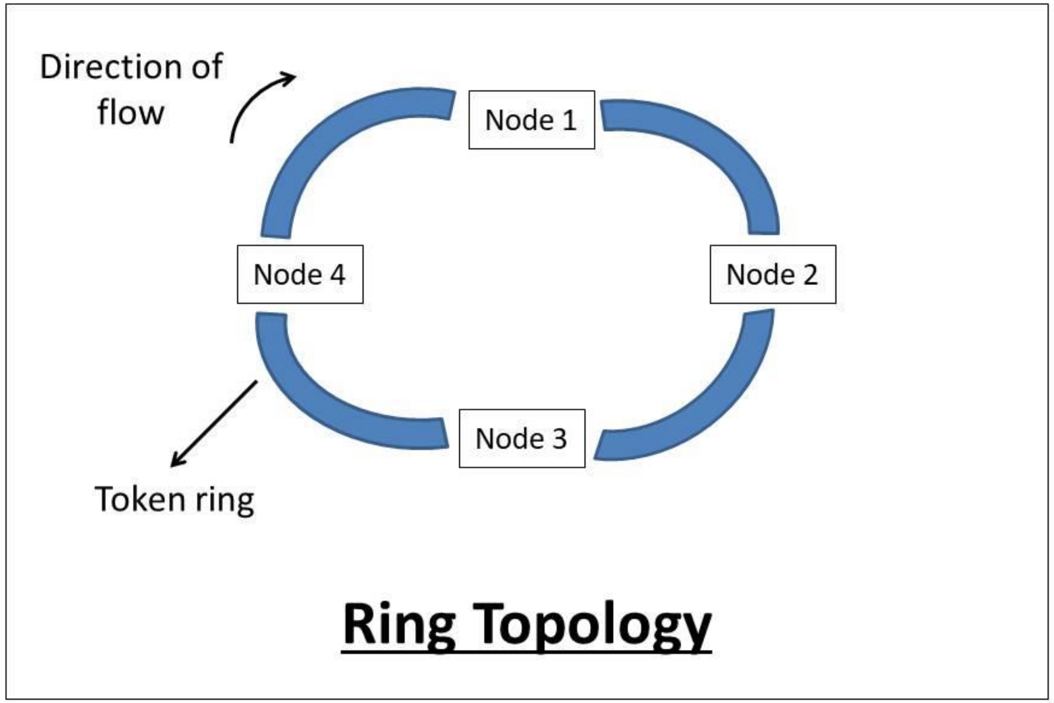
- Simple to use and install for small number of devices.
- If a node fails, it will not affect other nodes.
- Cost-efficient to implement due to shared media and less cabling required.

Disadvantages:

- If the bus fails, the network will fail. It’s a single point of failure
- When traffic is heavy, or nodes are too many, the performance significantly decreases.
- A limited number of nodes can connect to the bus due to limited bus length.
- Security issues and risks are more as messages are broadcasted to all nodes.
- Congestion and traffic on the bus as it is the only source of communication.

RING TOPOLOGY

- Each node is connected to two adjacent nodes to form the ring. The last node connected to the first forming a closed-loop.
- The message passing is unidirectional and circular in nature.
- Works in a token-based system and the token travels in a loop in one specific direction.
- If a token is free then the node can capture the token and attach the data and send to destination address.
- When this token reaches the destination node, the data is removed by the receiver and the token is made free to carry the next data.



RING TOPOLOGY

Advantages:

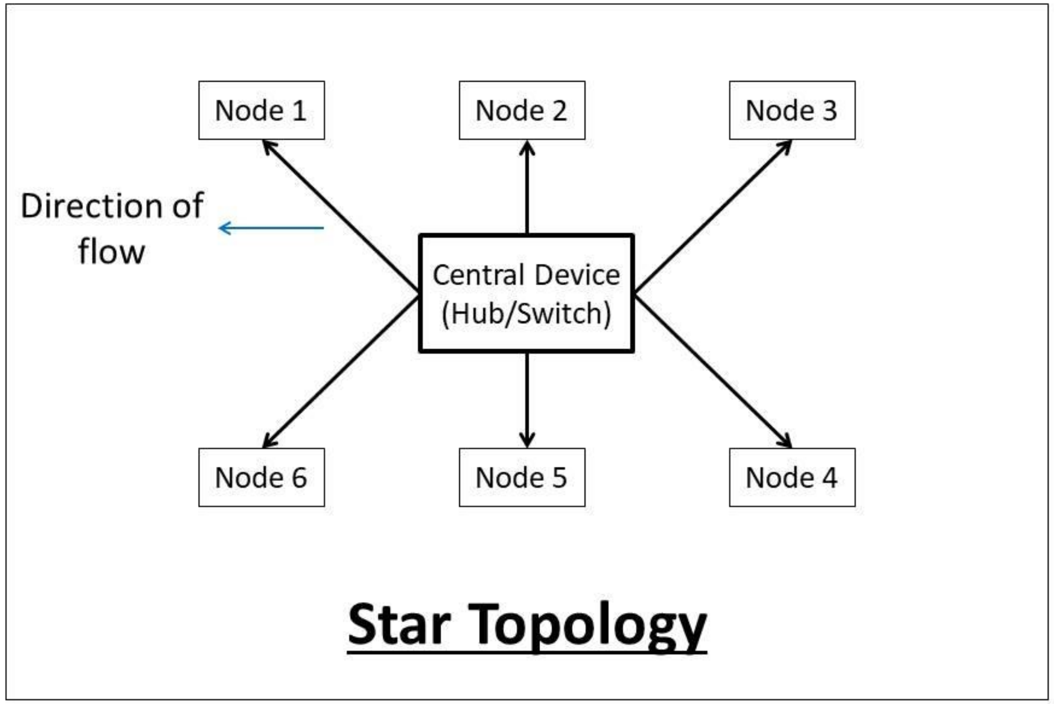
- Reduces chances of data collision (unidirectional).
- Easy to troubleshoot (the faulty node does not pass the token).
- Offers equal access to each node on the network

Disadvantages:

- If a node fails, the whole network will fail.
- Difficult to install and manage ring. Difficult to reconfigure (have to break ring).
- Slow data transmission speed (each message has to go through the ring path).

STAR TOPOLOGY

- All the nodes are connected to a centralized device (e.g. hub or switch)
- The central device acts as a server, and the other connected devices act as clients.
- The central device has point to point communication link (the dedicated link between the devices which can not be accessed by some other computer).
- Hubs tend to broadcast messages to all nodes, while the switches unicast the messages by maintaining a switching table.
- Switches are more recommended than hubs.



STAR TOPOLOGY

Advantages:

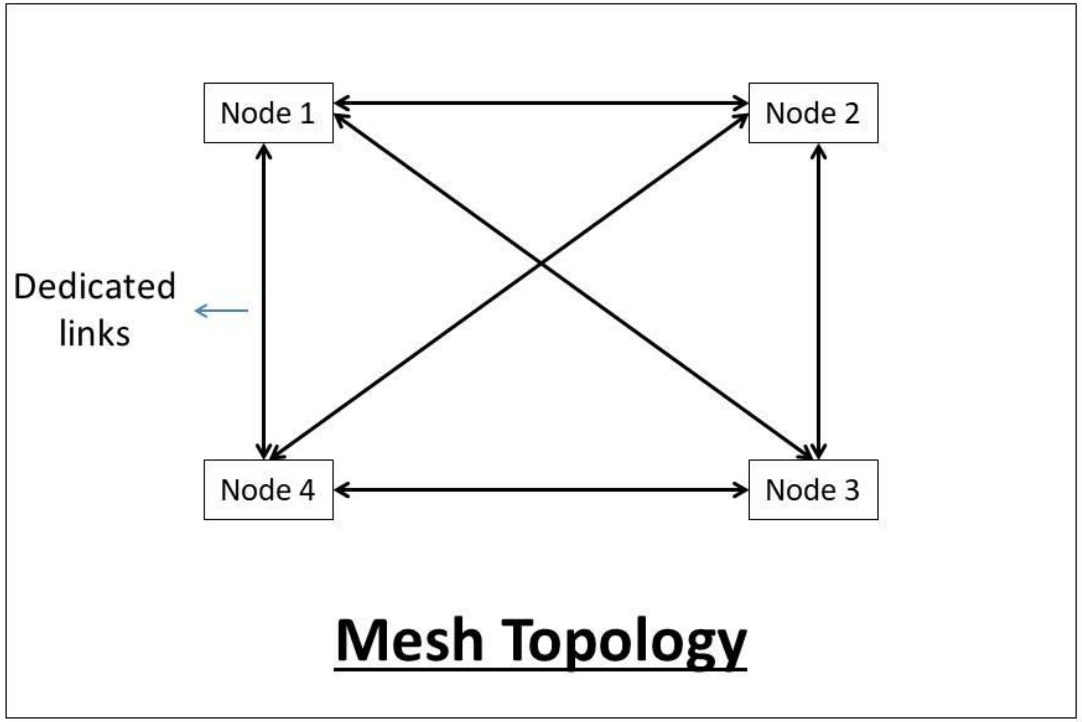
- Easy to scale (nodes can be added or removed to the network easily).
- Easy to troubleshoot (the faulty node does not give response).
- Good fault tolerance. If a node fails, it will not affect other nodes.
- Easy to reconfigure and upgrade due to centralized control (configured using a central device).

Disadvantages:

- If the central device fails, the network will fail. It is a single point of failure.
- The number of devices in the network is limited (due to limited input-output port in a central device).

MESH TOPOLOGY

- Nodes are interconnected with each other in a redundant fashion with multiple connections.
  - Full Mesh: Each node has a point to point link with every other node in the network.
  - Partial Mesh: Some nodes are not connected to every node in the network
- There are several different paths available in mesh topology.
- Direct communication can take place between different nodes in the network.
- The Internet is an example of partial mesh architecture



MESH TOPOLOGY

Advantages:

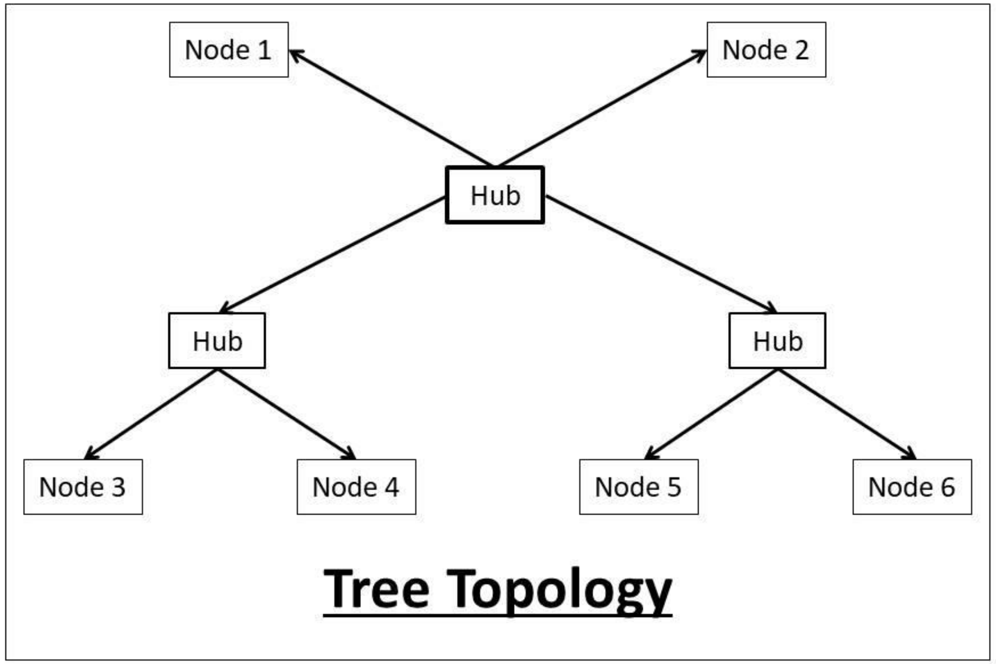
- Good fault tolerance. If a node fails, other alternatives are present in the network.
- Dedicated links facilitate direct and fast communication. No congestion or traffic problems on the channels.
- Maintains privacy and security due to a separate channel for communication.

Disadvantages:

- Very large amounts of cabling required.
- Installation and maintenance are very difficult.
- Cost inefficient to implement.
- Complex to implement and takes large space to install the network.

TREE/HIERARCHICHAL TOPOLOGY

- Has a root node, and all other nodes are connected in a hierarchy.
- There is a main/root hub/switch and all the other sub-hubs are connected to each other in this topology.
- Combination of Bus and Star topologies
- The whole network is divided into segments, which can be easily managed and maintained.



TREE TOPOLOGY

Advantages:

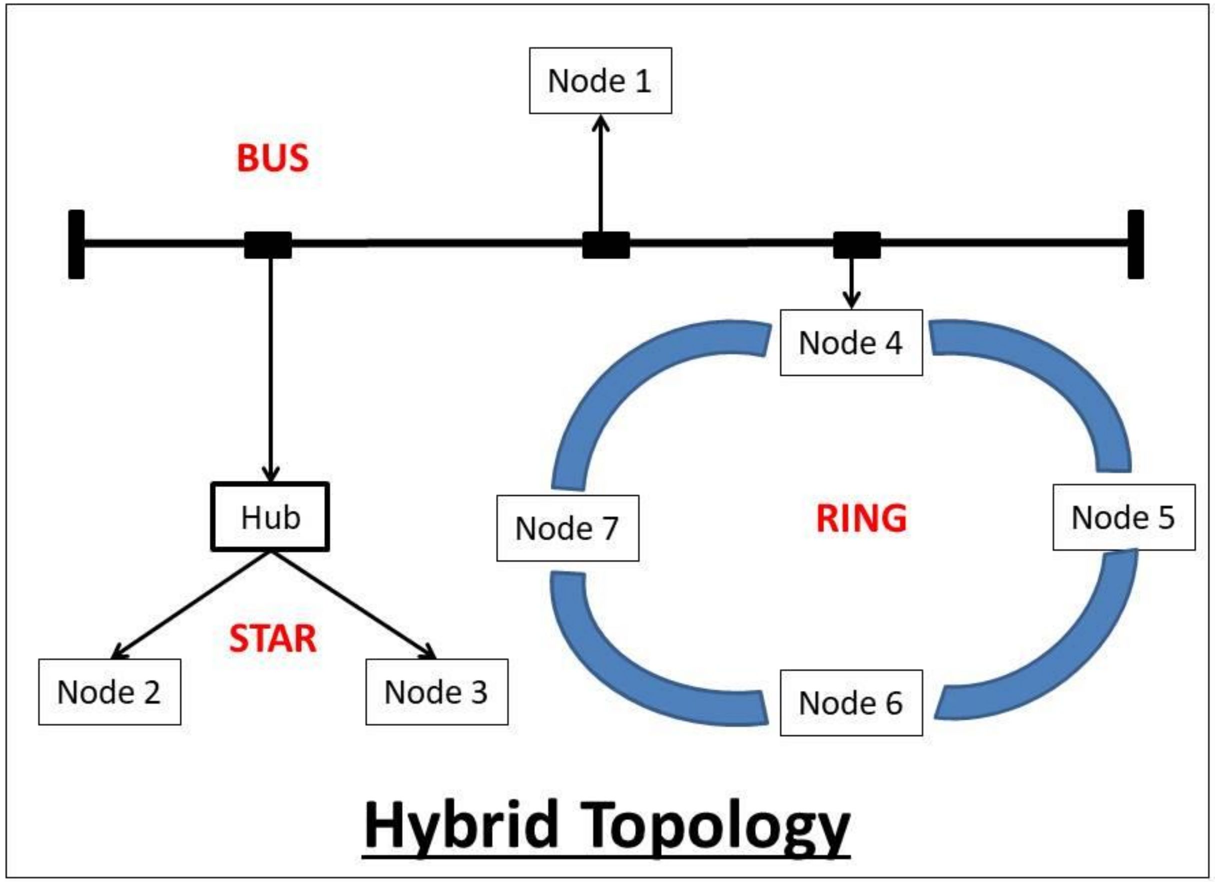
- Allows a large distance network coverage.
- A large number of nodes can be connected directly or indirectly.
- Fault finding is easy by checking each segment.
- Other hierarchical networks are not affected if one of them fails.

Disadvantages:

- If the main hub fails, the network will fail. It is a single point of failure
- Cabling and hardware cost is high. Complex to implement.
- A large network using tree topology can be hard to manage. It requires high maintenance.

HYBRID TOPOLOGY

- A combination of two or more topologies.
- Inherits the advantages and disadvantages of the topologies.
- All the good features of each topology can be used to make an efficient hybrid topology.



HYBRID TOPOLOGY

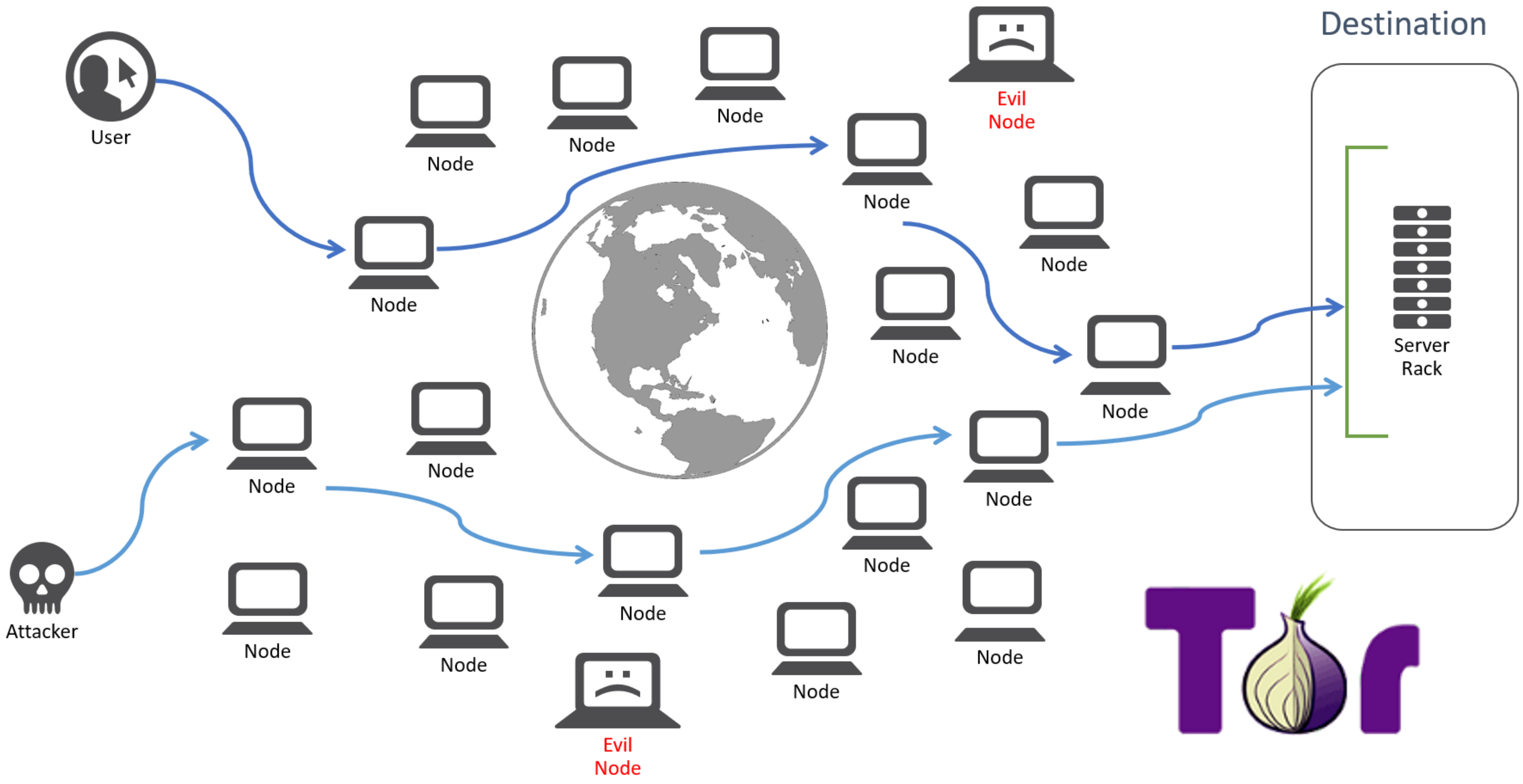
Advantages:

- It can handle a large volume of nodes.
- Reliable (if one node fails it will not affect the whole network).
- It provides flexibility to modify the network according to needs.

Disadvantages:

- Complex design.
- Expensive to implement.

The Onion Router (TOR) Network

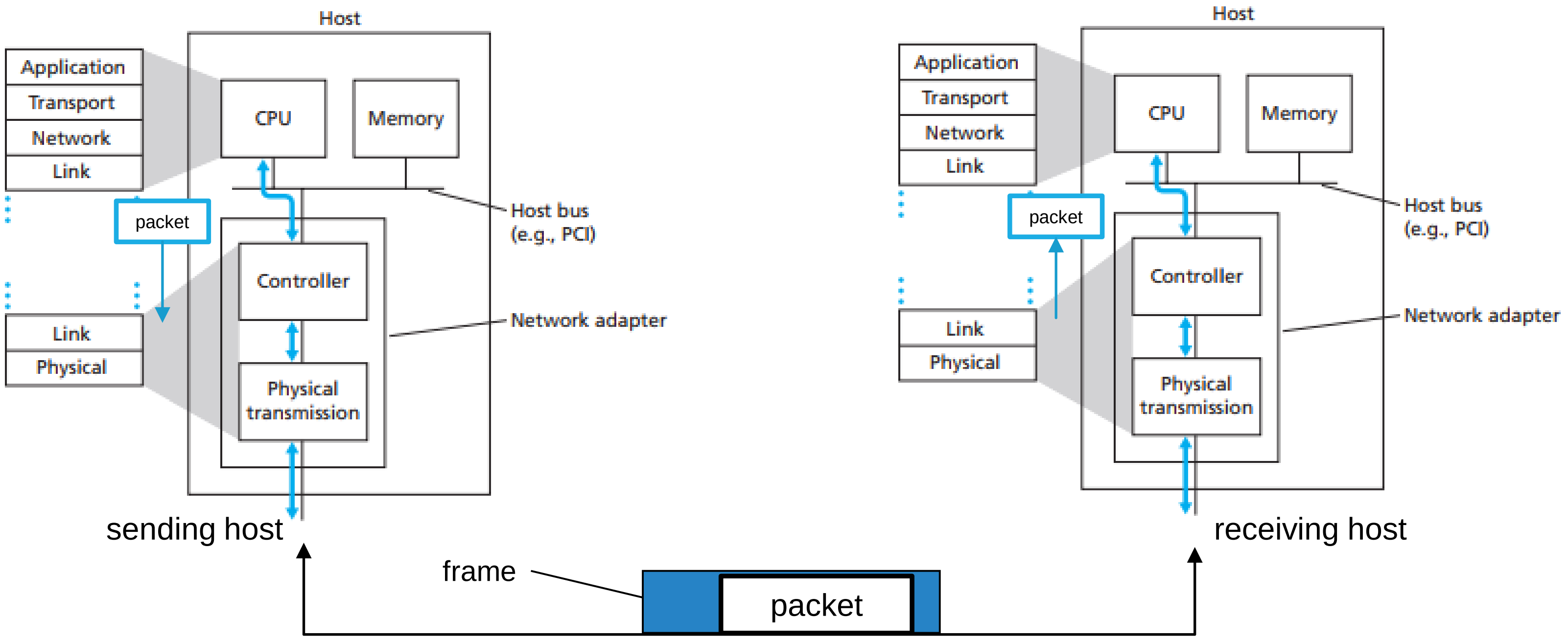


NETWORK INTERFACE & MEDIA ACCESS CONTROL

NETWORK INTERFACE CARD (NIC)

- Link layer is implemented in "adaptor" (aka network interface card NIC) e.g. Ethernet card, PCMCI card, 802.11 card
- Implements both data link and physical layer
- Attaches into host's system buses
- Combination of hardware, software, and firmware

IMPLEMENTATION



Sending side:

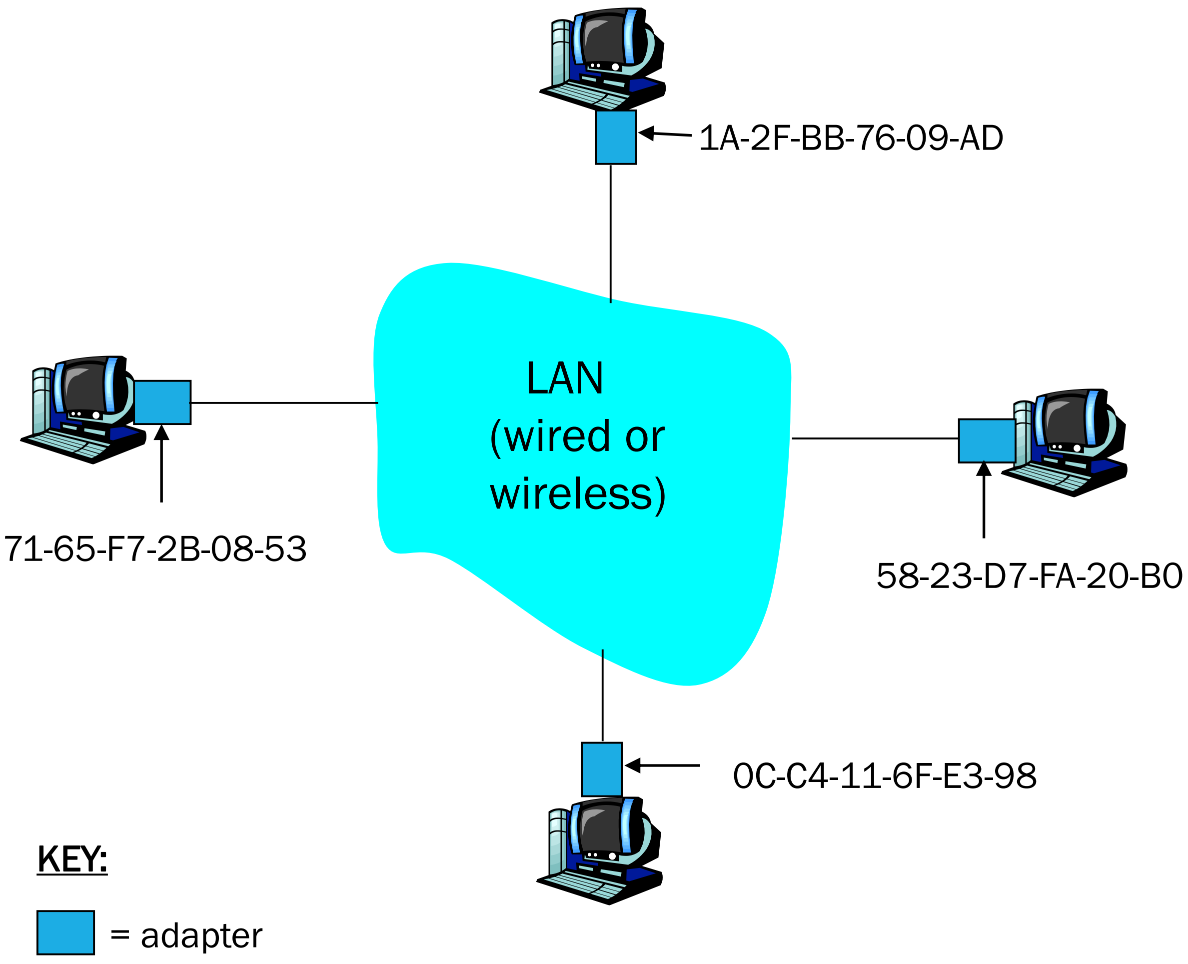
- Encapsulates datagram in frame
- Adds error checking bits, reliable delivery, flow control, etc.

Receiving side:

- Looks for errors, reliable delivery, flow control, etc.
- Extracts datagram, passes to upper layer

MAC ADDRESSES

- MAC (aka LAN or physical or Ethernet) address:
  - Function: get frame from one interface to another physically-connected interface (same network)
  - 48-bit (6 Bytes) MAC address
  - Burned in NIC ROM, also often software settable
- NB: IP address is a Network-layer address



Broadcast address =
FF-FF-FF-FF-FF-FF

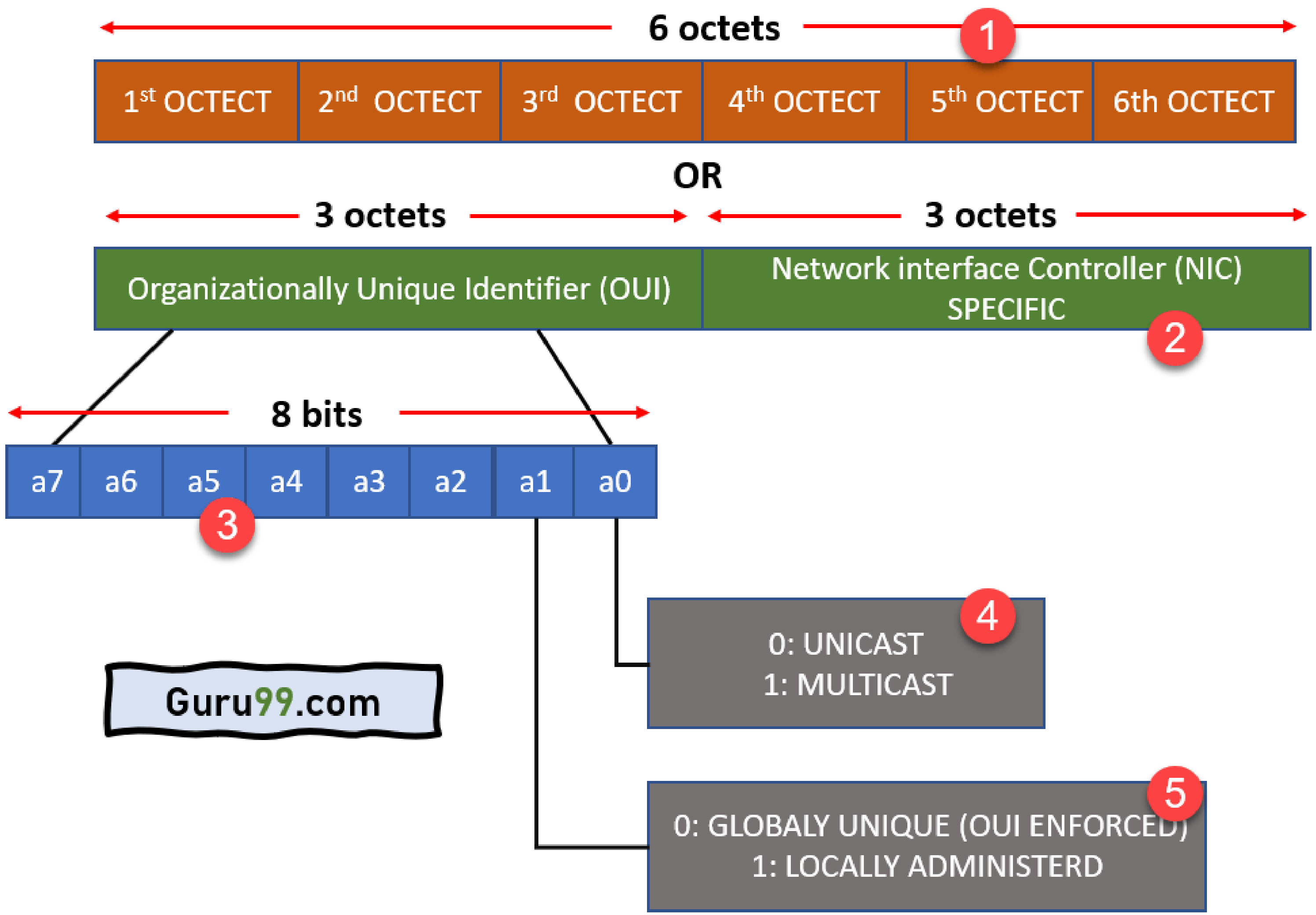
MAC ADDRESSES

- MAC address allocation administered by IEEE
- Manufacturer buys portion of MAC address space (to assure uniqueness)
- MAC flat address and has portability
  - Can move LAN card from one LAN to another
  - Akin to your national ID number
- IP hierarchical address NOT portable
  - Address depends on IP subnet to which node is attached
  - Akin to your postal address

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Example: Converting **168** to hex
Use the three-step process:
168 in binary is **10101000.**
In two groups of four binary digits is **1010** and **1000.**
1010is hex **A** and **1000** is hex **8.**
Answer: 168 is **A8** in hexadecimal.

MAC ADDRESSES



CLASS EXERCISE

Determine the MAC Address of your Host machine

- Right-click on the Start button and select Command Prompt.
- Enter the ipconfig /all command at the command prompt.
- How many MAC addresses did you discover in your PC? List all of them
- List the 3-byte Manufacturer OUI and the 3-byte Unique Interface Identifier for each MAC address
- Fill in the table below e.g.

Manufacturer OUI	Unique Interface Identifier	Vendor Name
D4-BE-D9	13-63-00	Dell Incorporated

ADDRESS RESOLUTION PROTOCOL (ARP)

ADDRESS RESOLUTION PROTOCOL (ARP)

- ARP was defined in 1982 by RFC 826.
- ARP is a communication protocol used to discover the Data-Link Layer address e.g. MAC address associated with an Internet layer address e.g. IPv4 address.
- ARP is a request-response or request-reply protocol
- ARP is needed because devices in a LAN e.g. switches are programmed to communicate using link layer addresses.
- Every device that is capable of handling IPv4 packets has an ARP table which consists of IPv4 address to MAC address mappings.

ADDRESS RESOLUTION PROTOCOL (ARP) TABLE

Neighbour	Linklayer Address	Expire (O)	Expire (I) Netif	Refs	Prbs
10.0.0.1	0:c:29:12:34:56	2m25s	2m25s	en0	1 none
10.0.0.41	0:c:29:65:43:21	39s	39s	en0	1 none
10.0.0.47	0:c:00:00:00:01	1m14s	1m14s	en0	1 none
10.0.0.53	ab:cd:ef:81:6:72	2m32s	2m32s	en0	1 none
10.0.0.58	fe:dc:ba:11:e2:6e	1m51s	1m51s	en0	1 none

- Neighbor: The IP address of another device connected to the same network.
- Link layer address: The MAC address of the device connected to the same network.
- Expire: A timer, counting down until the specific entry is no longer considered valid and is flushed (removed) from the ARP table.
- Netif: The specific interface that a MAC address was discovered on.

ADDRESS RESOLUTION PROTOCOL (ARP)

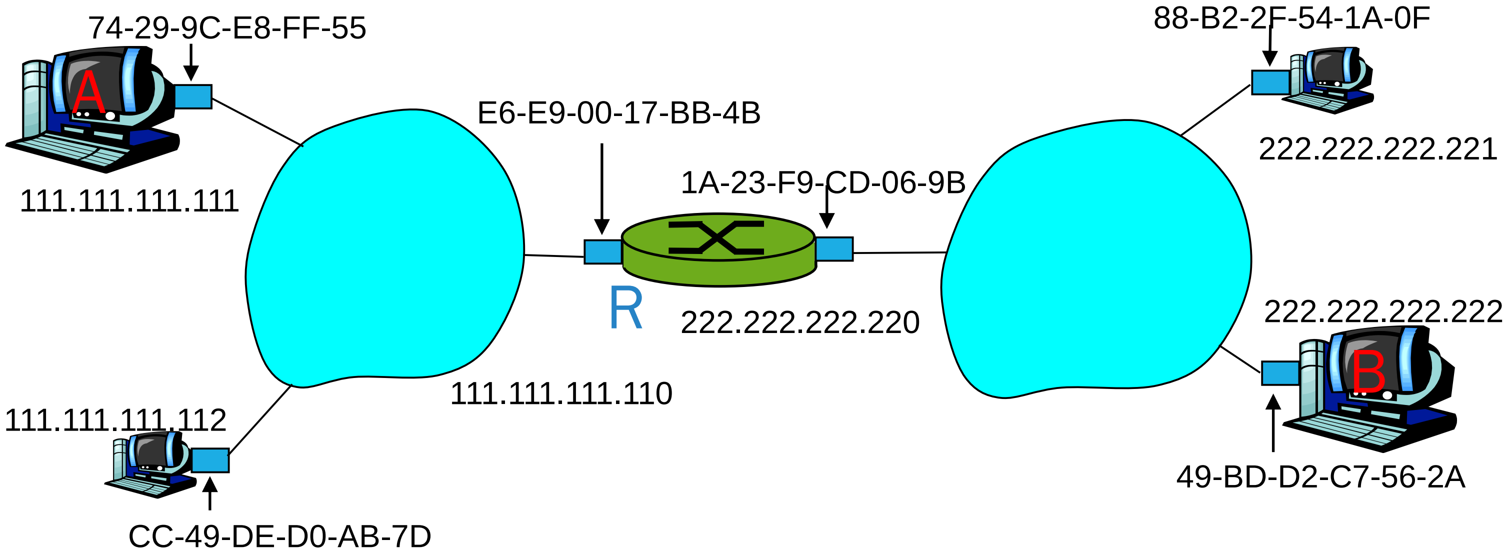
- Each IP node (host) on LAN has ARP table
- ARP table is built from the replies to the ARP requests
  - Has IP address - MAC address mapping and TTL
  - TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)
- Switches maintain a MAC address table aka MAC Forwarding Table or Forwarding Database (FDB), mapping information on the physical switch port a specific device is connected to i.e. the port where packets should go to reach that device.
- ARP tables map a Layer 3 address to a Layer 2 address configuration, while MAC tables map a Layer 2 address to a Layer 1 (physical layer) interface.

ARP PROTOCOL: SAME LAN

- A wants to send frame to B, and B’s MAC address not in A’s ARP table.
- A broadcasts ARP query packet, containing B's IP address
  - Dest MAC address = FF-FF-FF-FF-FF-FF
  - all machines on LAN receive ARP query
- B receives ARP packet, replies to A with its (B's) MAC address
  - frame sent to A’s MAC address (unicast)
- A caches (saves) IP-to-MAC address pair in its ARP table until information becomes old (times out)
  - soft state: information that times out (goes away) unless refreshed
- ARP is “plug-and-play”:
  - nodes create their ARP tables *without intervention from net administrator*

ROUTING TO ANOTHER LAN

- Here there are two ARP tables in router R, one for each IP network (LAN)



- send data from A to B via R. Assume A knows B’s IP address

ROUTING TO ANOTHER LAN

- A creates frame with source A, destination B
- A uses ARP to get R’s MAC address for 111.111.111.110
- A creates link-layer frame with R's MAC address as dest, frame contains A-to-B IP datagram
- A’s NIC sends frame
- R’s NIC receives frame
- R removes IP packet from Ethernet frame, sees its destined to B
- R uses ARP to get B’s MAC address
- R creates frame containing A-to-B IP packet sends to B

TYPES OF ARP

Proxy ARP

- Proxy ARP is a technique by which a proxy device on a given network answers the ARP request for an IP address that is not on that network. The proxy is aware of the location of the traffic's destination and offers its own MAC address as the destination.

Gratuitous ARP

- Gratuitous ARP is not prompted by an ARP request to translate an IP address to a MAC address.
- Gratuitous ARP is almost like an administrative procedure, carried out as a way for a host on a network to simply announce or update its IP-to-MAC address.

Reverse ARP (RARP)

- Host machines that do not know their own IP address can use the Reverse Address Resolution Protocol (RARP) for discovery.

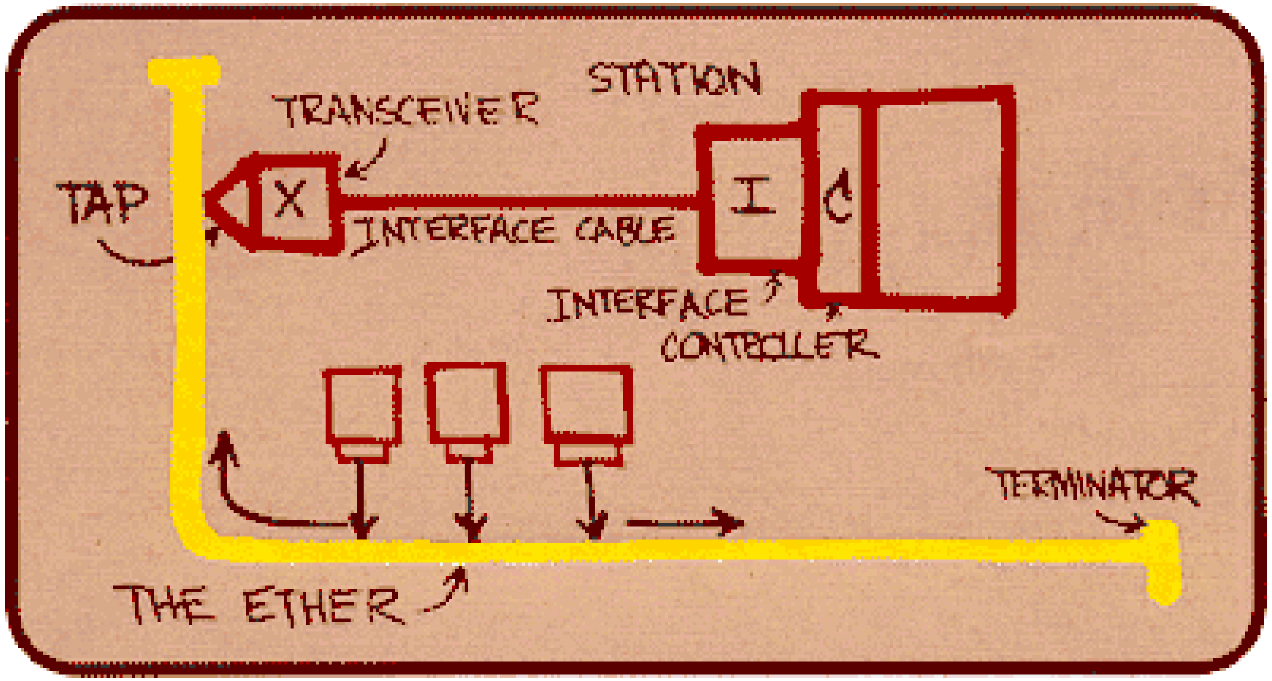
Inverse ARP (IARP)

- Whereas ARP uses an IP address to find a MAC address, IARP uses a MAC address to find an IP address.

ETHERNET

ETHERNET

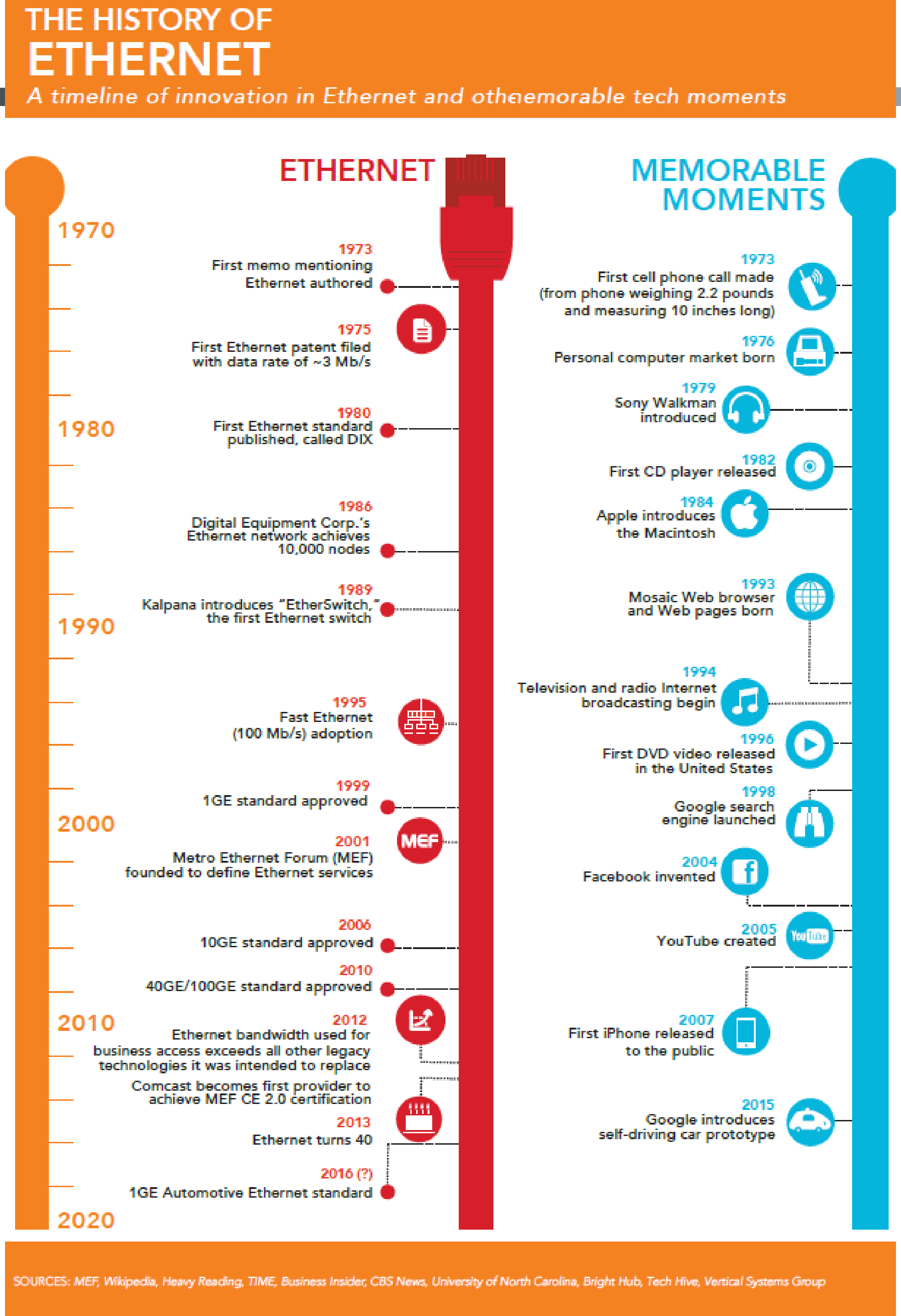
- On May 22, 1973, Robert Metcalfe, by the time PhD student in electrical engineering working at Xerox Palo Alto Research Center (PARC), wrote a memo titled "Alto Ethernet" describing a way to transmit data from the early generation of personal computers to a new device, the laser printer.
- November 11, 1973 is the first day the system actually functioned

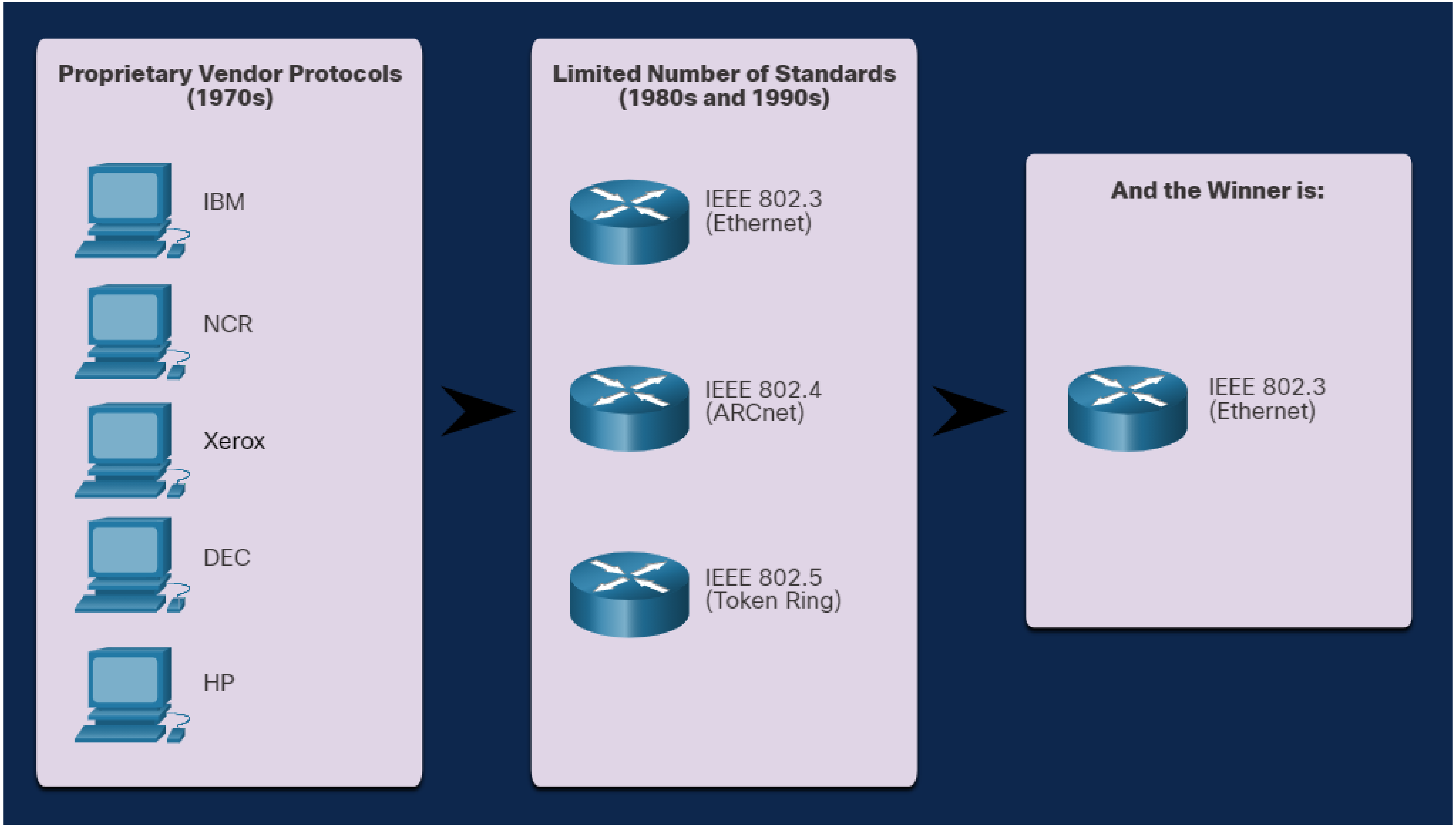


Sketch drawn by Robert M. Metcalfe in 1976 to present Ethernet to the National Computer Conference in June of that year.

ETHERNET

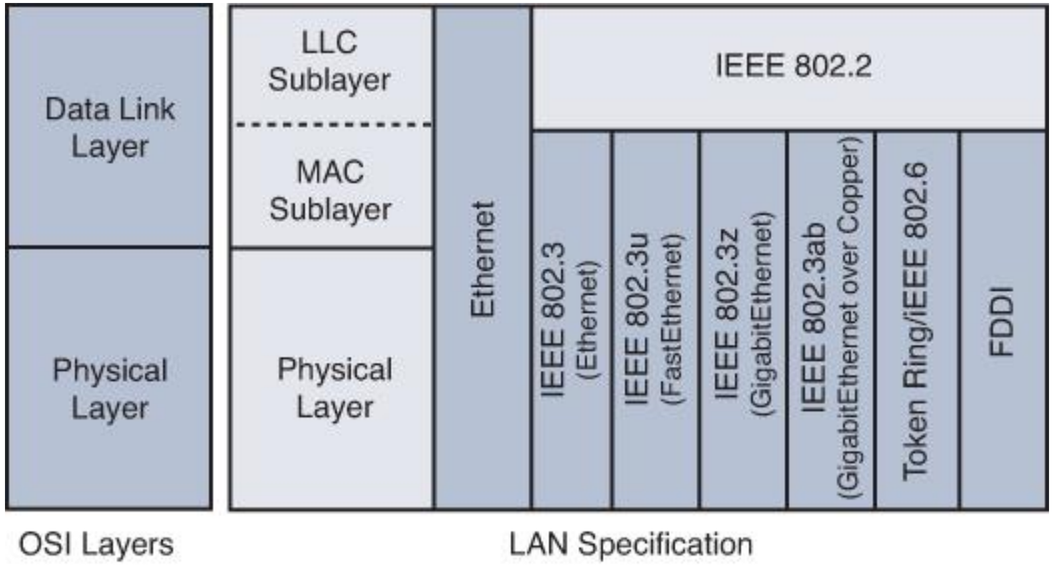
- Created in 1973. The committee responsible for the Ethernet standards is 802.3
- It has evolved over the years. Each version of Ethernet has an associated standard e.g. 100BASE-T represents the 100 Megabit Ethernet using twisted-pair cable standards. The standard notation translates as:
  - 100 is the speed in Mbps
  - BASE stands for baseband transmission
  - T stands for the type of cable, in this case, twisted-pair.
- Early versions were relatively slow at 10 Mbps. The latest versions operate at 10 Gigabits per second and more


THE HISTORY OF
ETHERNET
A timeline of innovation in Ethernet and othaemorable tech moments
ETHERNET
MEMORABLE MOMENTS
1970
1980
1990
2000
2010
2020
1973
First memo mentioning Ethernet authored
1975
First Ethernet patent filed with data rate of ~3 Mb/s
1980
First Ethernet standard published, called DIX
1986
Digital Equipment Corp.'s Ethernet network achieves 10,000 nodes
1989
Kalpana introduces "EtherSwitch," the first Ethernet switch
1995
Fast Ethernet (100 Mb/s) adoption
1999
1GE standard approved
MEF
2001
Metro Ethernet Forum (MEF) founded to define Ethernet services
2006
10GE standard approved
2010
40GE/100GE standard approved
2012
Ethernet bandwidth used for business access exceeds all other legacy technologies it was intended to replace
Comcast becomes first provider to achieve MEF CE 2.0 certification
2013
Ethernet turns 40
2016 (?)
1GE Automotive Ethernet standard
1973
First cell phone call made (from phone weighing 2.2 pounds and measuring 10 inches long)
1976
Personal computer market born
1979
Sony Walkman introduced
1982
First CD player released
1984
Apple introduces the Macintosh
1993
Mosaic Web browser and Web pages born
1994
Television and radio Internet broadcasting begin
1996
First DVD video released in the United States
1998
Google search engine launched
2004
Facebook invented
2005
YouTube created
2007
First iPhone released to the public
2015
Google introduces self-driving car prototype
SOURCES: MEF, Wikipedia, Heavy Reading, TIME, Business Insider, CBS News, University of North Carolina, Bright Hub, Tech Hive, Vertical Systems Group


Proprietary Vendor Protocols (1970s)
IBM
NCR
Xerox
DEC
HP
Limited Number of Standards (1980s and 1990s)
IEEE 802.3 (Ethernet)
IEEE 802.4 (ARCnet)
IEEE 802.5 (Token Ring)
And the Winner is:
IEEE 802.3 (Ethernet)

ETHERNET

- Many different Ethernet standards
- Common MAC protocol and frame format
- Different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1Gbps, 10G bps
- Different physical layer media: fiber/copper cable



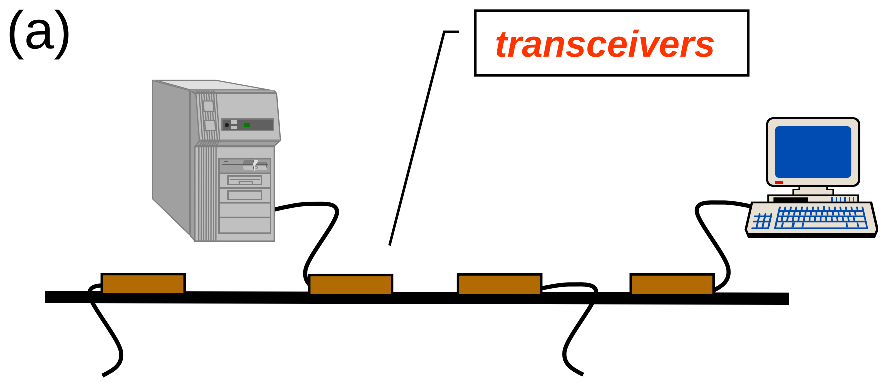
ETHERNET

- Star topology prevails. Active switch in center (contrast with hub). Each “spoke” runs separate (nodes do not collide with each other)
- Baseband signals broadcast
  - Baseband refers to the original frequency range of a transmission signal before it is modulated. Also refers to a type of data transmission in which digital or analog data is sent over a single non-multiplexed channel.
- Uses Manchester Encoding
- 32-bit Cyclic Redundancy Check (CRC) for error detection
- CSMA-CD with Binary Exponential Backoff
- Single segments up to 500m; with up to 4 repeaters gives 2500m max length
- Max 100 stations/segment, 1024 stations/Ethernet

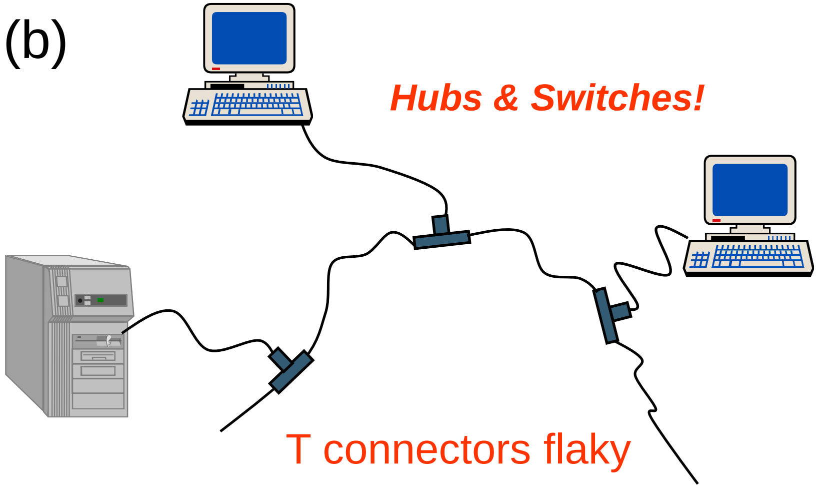
IEEE 802.3 PHYSICAL LAYER

	10base5	10base2	10baseT	10baseFX
Medium	Thick coax	Thin coax	Twisted pair	Optical fiber
Max. Segment Length	500 m	200 m	100 m	2 km
Topology	Bus	Bus	Star	Point-to-point link

IEEE 802.3 10 Mbps Medium Alternatives



Thick Coax: Stiff, hard to work with



T connectors flaky

FAST ETHERNET

	100baseT4	100baseT	100baseFX
Medium	Twisted pair category 3 UTP 4 pairs	Twisted pair category 5 UTP two pairs	Optical Fiber Multimode Two strands
Max. Segment Length	100 m	100 m	2 km
Topology	Star	Star	Star

IEEE 802.3 100 Mbps Ethernet Medium Alternatives

Fast Ethernet introduced in 1995 and is covered under 802.3u

To preserve compatibility with 10 Mbps Ethernet:

- Same frame format, same interfaces, same protocols
- Bus topology & coaxial cable abandoned
- Hub topology only with twisted pair & fiber

GIGABIT ETHERNET

	1000baseSX	1000baseLX	1000baseCX	1000baseT
Medium	Optical Fiber Multimode Two strands	Optical Fiber Single Mode Two strands	Shielded copper cable	Twisted pair Category 5, 5e, 6 and 7 UTP
Max. Segment Length	550 m	5 km	25 m	100 m
Topology	Star	Star	Star	Star

IEEE 802.3 1 Gbps Fast Ethernet Medium Alternatives

Gigabit Ethernet introduced in 1999 as 802.3z for fiber and 802.3ab for copper

- Minimum frame length increased to *512 Bytes*
- Frame bursting to allow stations to transmit burst of short frames
- Frame structure preserved but CSMA-CD essentially abandoned

10 GIGABIT ETHERNET

	10GbaseSR	10GBaseLR	10GbaseEW	10GbaseLX4
Medium	Two optical fibers Multimode at 850 nm 64B66B code	Two optical fibers Single-mode at 1310 nm 64B66B	Two optical fibers Single-mode at 1550 nm SONET compatibility	Two optical fibers multimode/single-mode with four wavelengths at 1310 nm band 8B10B code
Max. Segment Length	300 m	10 km	40 km	300 m – 10 km

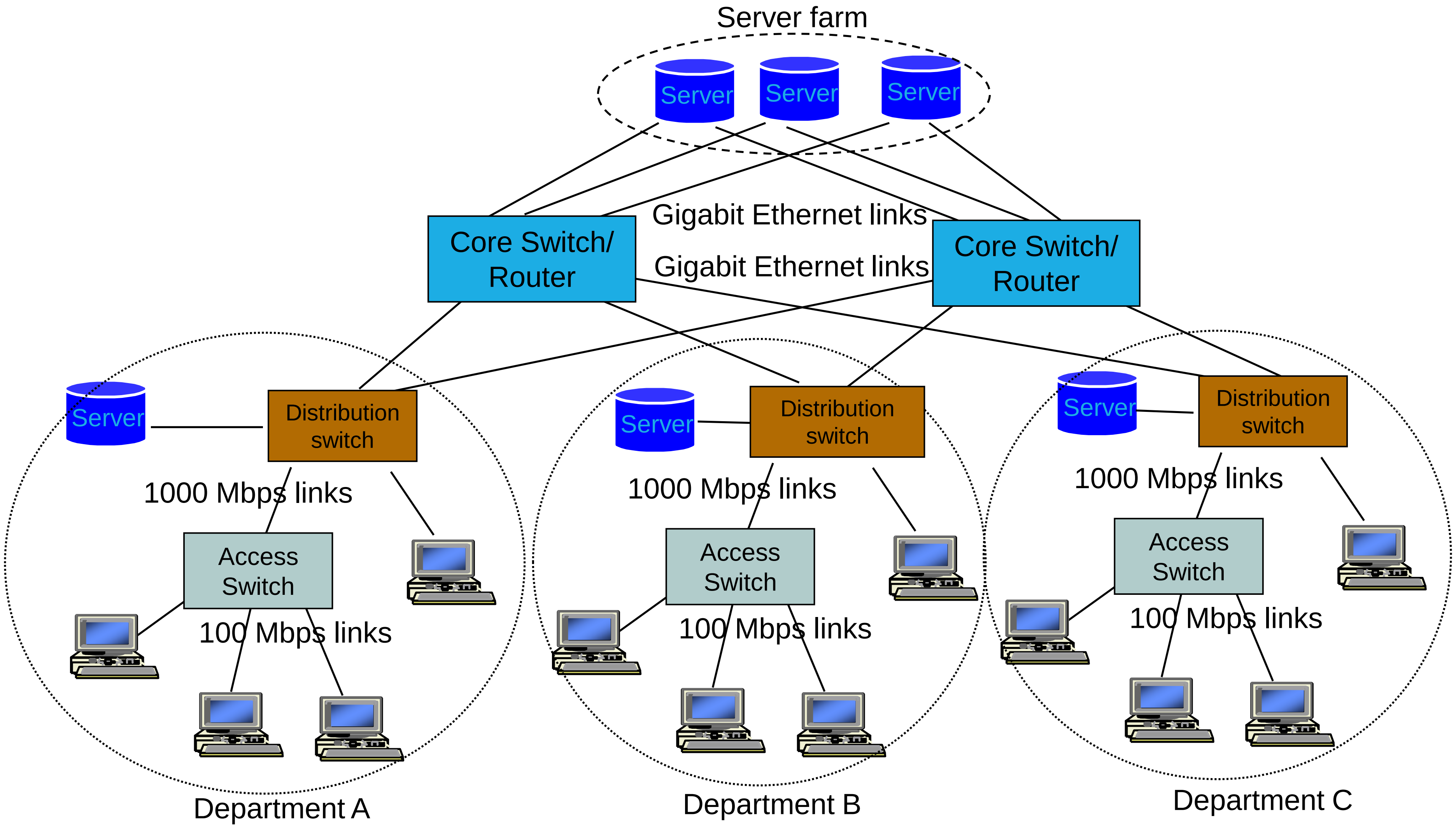
IEEE 802.3 10 Gbps Ethernet Medium Alternatives

- First defined by the IEEE 802.3ae standard in 2002 .
- Only supports full-duplex point-to-point links connected by network switches
- CSMA-CD protocol officially abandoned
- Can use either copper or fiber cabling. Different physical (PHY) layer modules may be plugged.
- Maximum distance over copper cable is 100 meters but because of its bandwidth requirements, higher-grade copper cables are required.

ETHERNET STANDARDS

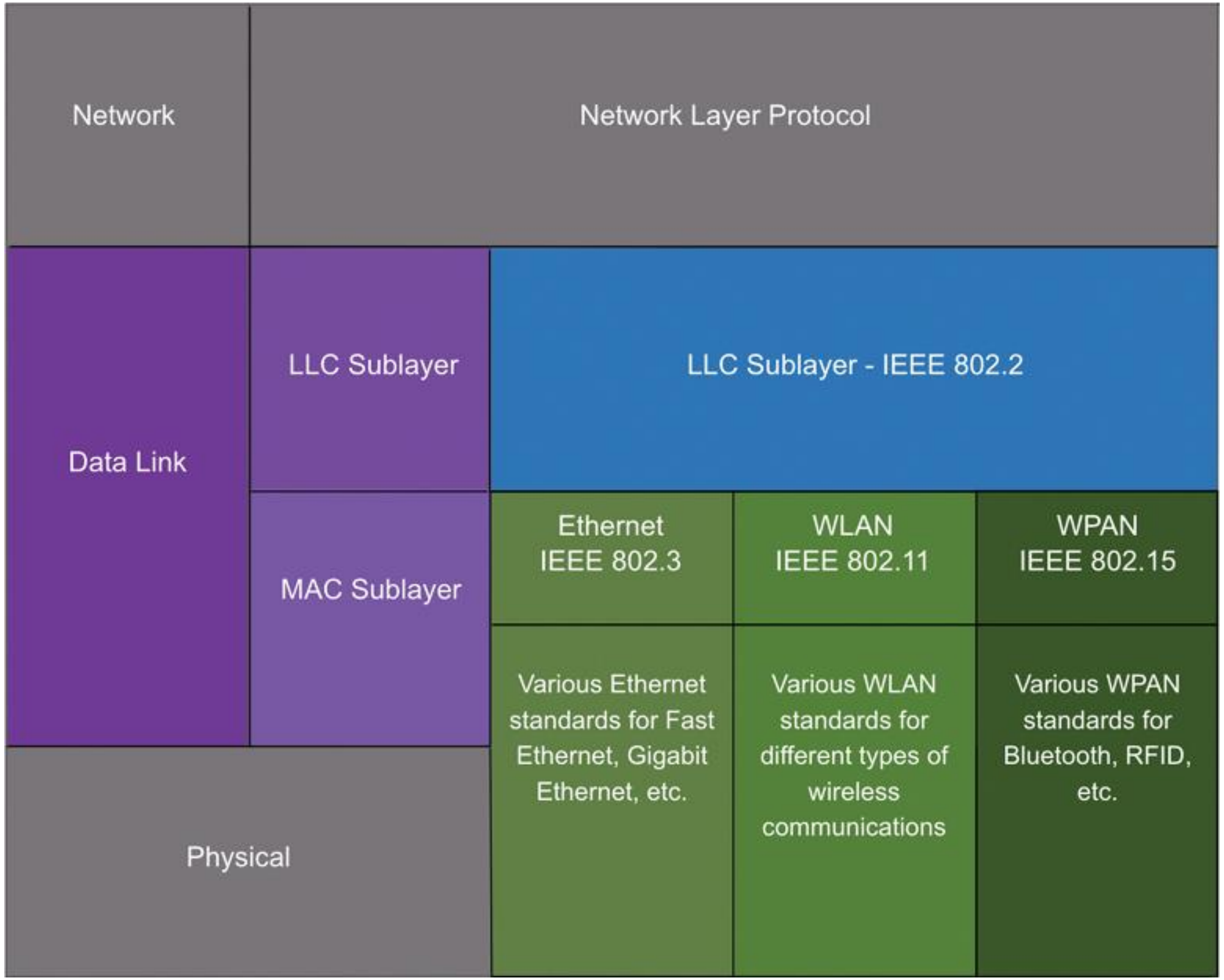
Name	Code	Standard	Data	Distance	Cable
Ethernet	10BASE-T	802.3i-1990	10 Mbits	100 m	Copper
Fast Ethernet	100BASE-T	802.3u-1995	100 Mbits	100 m	Copper
Fast Ethernet	100BASE-SX	802.3u-1995	100 Mbits	2000 m	Fibre
Giga Ethernet	1000BASE-T	802.3ab-1999	1000 Mbits	100 m	Copper
Giga Ethernet	1000BASE-LX	802.3z-1998	1000 Mbits	5 km	Fibre
10 Gigabit Ethernet	10GBASE-T	802.3an-2006	10 Gbits	100 m	Copper
10 Gigabit Ethernet	10GBASE-LR	802.3ae-2002	10 Gbits	10 km	Fibre
100 Gigabit Ethernet	100GBASE-LR4	802.3ba-2010	100 Gbits	10 km	Fibre
Terabit Ethernet	400GBASE-LR8	802.3bs-2017	400 Gbits	10 km	Fibre

TYPICAL ETHERNET DEPLOYMENT



ETHERNET FRAMES & OPERATIONS

SUBLAYERS


Network
Network Layer Protocol
Data Link
LLC Sublayer
LLC Sublayer - IEEE 802.2
MAC Sublayer
Ethernet
IEEE 802.3
WLAN
IEEE 802.11
WPAN
IEEE 802.15
Various Ethernet standards for Fast Ethernet, Gigabit Ethernet, etc.
Various WLAN standards for different types of wireless communications
Various WPAN standards for Bluetooth, RFID, etc.
Physical

SUBLAYERS

According to IEEE 802, the data link layer is often divided into two sublayers: logical link control (LLC) and media access control (MAC).

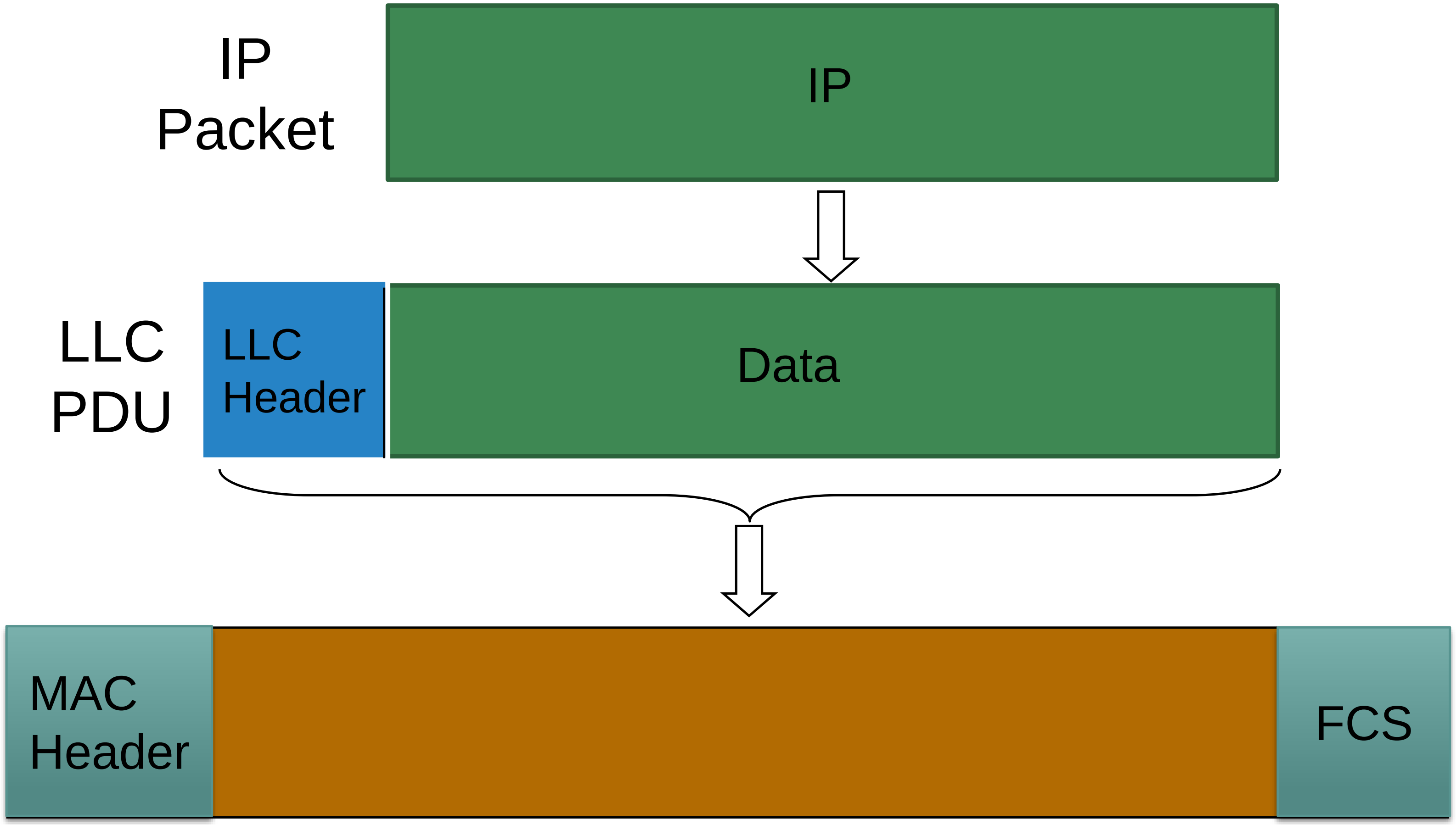
Logical Link Control sublayer

- Provides services to upper layers e.g. flow control, acknowledgment, and error notification.

Media Access Control sublayer

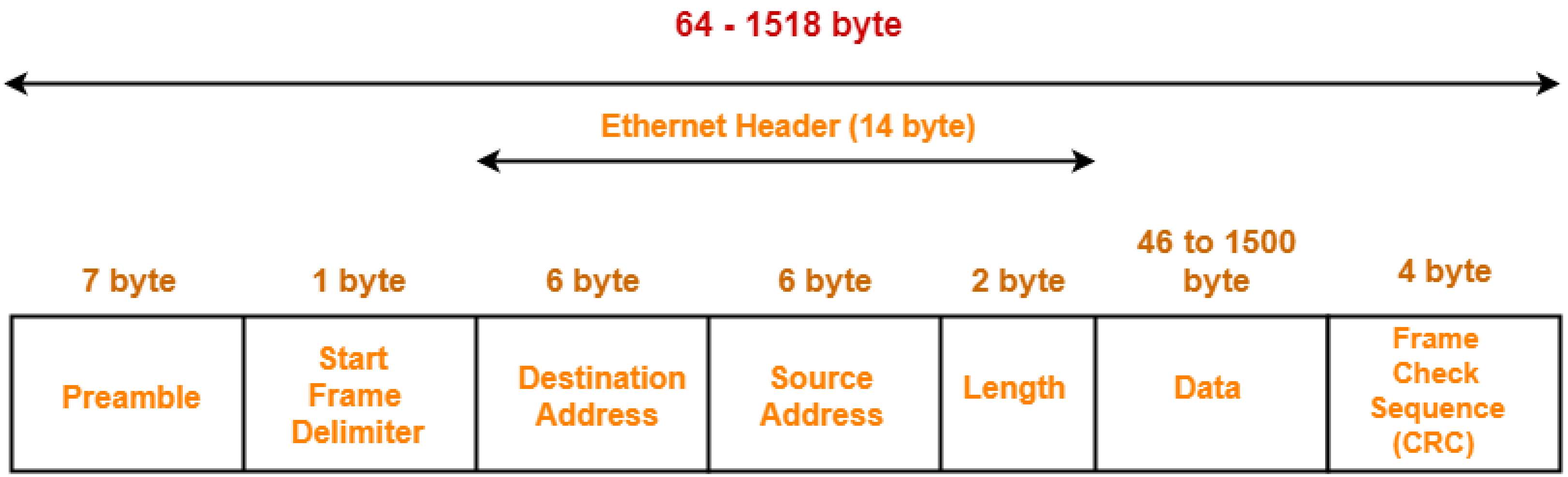
- Determines who is allowed to access the media at any one time (e.g. CSMA/CD). Other times it refers to a frame structure delivered based on MAC addresses inside.
- Also performs frame synchronization, which determines the start and end of each frame of data in the transmission bitstream.

ENCAPSULATION



ETHERNET FRAME STRUCTURE

- Sending adapter encapsulates IP packet (or other network layer protocol packet) in an Ethernet frame



IEEE 802.3 Ethernet Frame Format

ETHERNET FRAME STRUCTURE

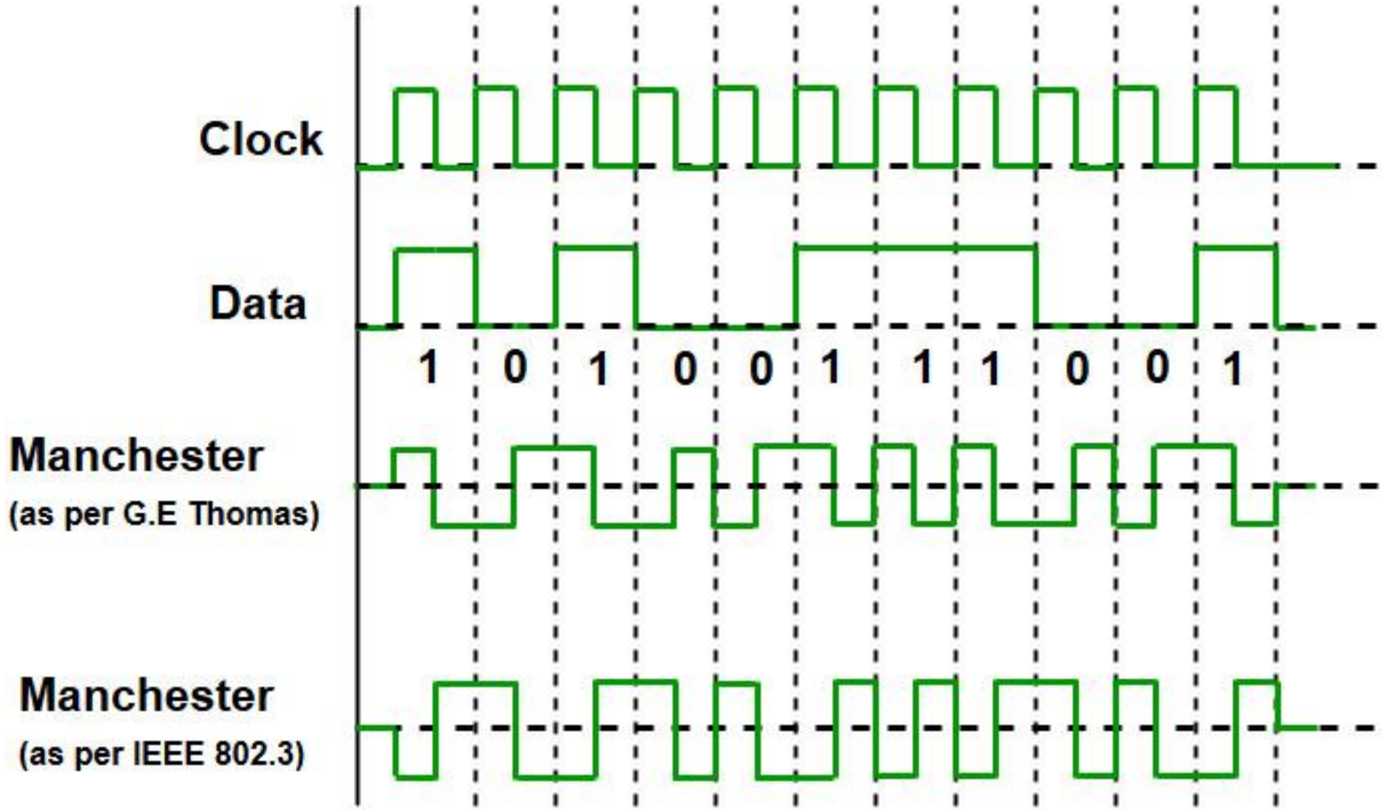
- Preamble: 7 Bytes with pattern 10101010. Used to synchronize receiver, sender clock rates. Allows the receiver to lock onto the data stream before the actual frame begins.
- Start of Frame Delimiter (SFD): This is a 1-Byte field which is always set to 10101011. Sometimes SFD is considered the part of PRE, this is the reason Preamble is described as 8 Bytes in some places. This is the last chance for synchronization.
- Destination Address: This is 6-Byte field which contains the MAC address of machine for which data is destined.
  - If matching destination or broadcast address (e.g. ARP packet), it passes data in frame to network layer protocol. Otherwise, adapter discards frame.
- Source Address: This is a 6-Byte field which contains the MAC address of source machine. It is always an individual Unicast address - the least significant bit of first byte is always 0

ETHERNET FRAME STRUCTURE

- Length: A 2-Byte field, which indicates the length of entire Ethernet frame. A 16-bit field can hold value 0 to 65534, but length cannot exceed 1500 due to limitations of Ethernet.
- Data: Actual data is inserted here, aka. Payload. Both IP header and data inserted if IP is used over Ethernet. The Ethernet maximum data is 1500 Bytes and minimum is 46 bytes (padded with 0s to meet minimum)
- Cyclic Redundancy Check (CRC): 4 Byte field i.e. contains a 32-bits hash code of Destination Address, Source Address, Length, and Data fields. If the checksum computed by destination is not the same as sent checksum value, data received is corrupted.

MANCHESTER ENCODING

- Used in 10BaseT
- Each bit has a transition
- Allows clocks in sending and receiving nodes to synchronize to each other
  - No need for a centralized, global clock among nodes!
- This is physical-layer function

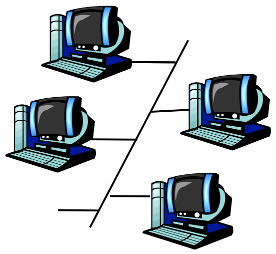


MEDIA ACCESS CONTROL

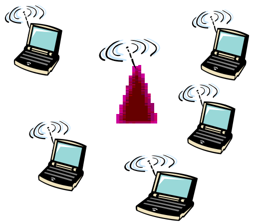
MULTIPLE ACCESS LINKS AND PROTOCOLS

Two types of links:

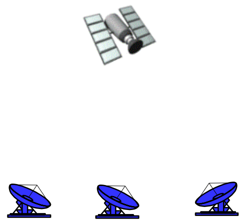
- Point-to-point (not shared)
  - PPP for dial-up access
  - point-to-point link between Ethernet switch and host
- Broadcast (shared wire or medium)
  - Old-fashioned Ethernet
  - 802.11 wireless LAN



shared wire (e.g., cabled Ethernet)



shared RF (e.g., 802.11 WiFi)



shared RF (satellite)

humans at a cocktail party (shared air, acoustical)

MULTIPLE ACCESS PROTOCOLS

- Single shared broadcast channel
- Two or more simultaneous transmissions by nodes
  - collision if node receives two or more signals at the same time
- Multiple Access protocol
- Distributed algorithm determines how nodes share channel (i.e. determine when/who node can transmit)
- Communication about channel sharing must use channel itself!
  - no “out-of-band” channel for coordination

MULTIPLE ACCESS PROTOCOLS

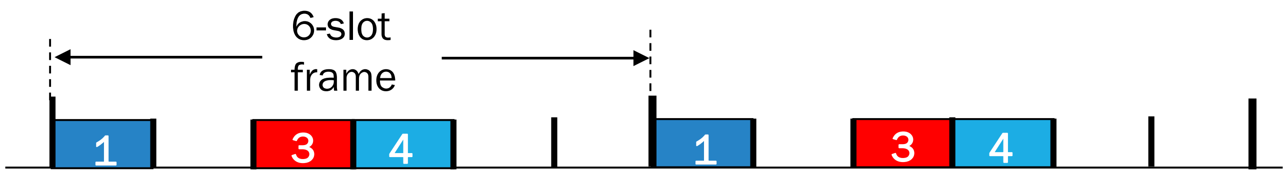
Three broad classes:

- Channel Partitioning
  - Divide channel into smaller “pieces” (time slots, frequency)
  - Allocate piece to node for exclusive use
- Random Access
  - Channel not divided, allow collisions
  - “Recover” from collisions
- Taking Turns
  - Nodes take turns, but nodes with more to send can perhaps take longer turns

CHANNEL PARTITIONING: TDMA

TDMA: Time Division Multiple Access

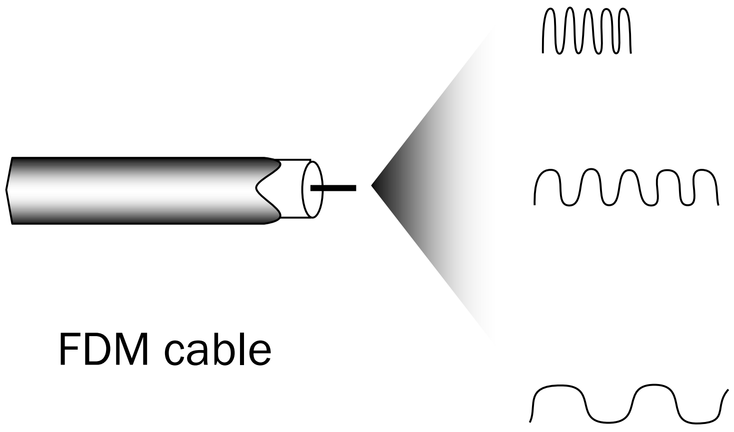
- Access to channel in "rounds"
- Each station gets fixed length slot (length = pkt trans time) in each round
- Unused slots go idle
- Example: 6-station LAN, 1,3,4 have pkt, slots 2,5,6 idle

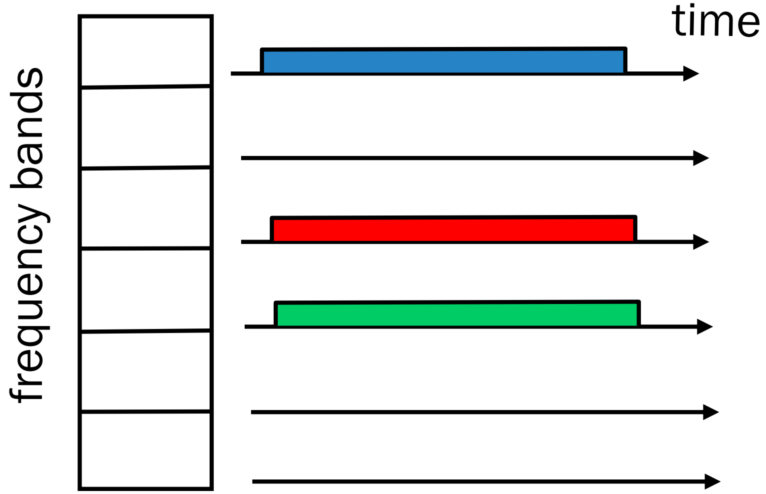


CHANNEL PARTITIONING: FDMA

FDMA: Frequency Division Multiple Access

- Channel spectrum divided into frequency bands
- Each station assigned fixed frequency band
- Unused transmission time in frequency bands go idle
- Example: 6-station LAN, 1,3,4 have pkt, frequency bands 2,5,6 idle





RANDOM ACCESS PROTOCOLS

- When node has packet to send
  - Transmit at full channel data rate R
  - No a priori coordination among nodes
- Two or more transmitting nodes results in collision
- Random Access MAC protocol specifies:
  - How to detect collisions
  - How to recover from collisions (e.g. via delayed retransmissions)
- Examples of Random Access MAC protocols are: ALOHA, CSMA, CSMA/CD, CSMA/CA

CSMA (CARRIER SENSE MULTIPLE ACCESS)

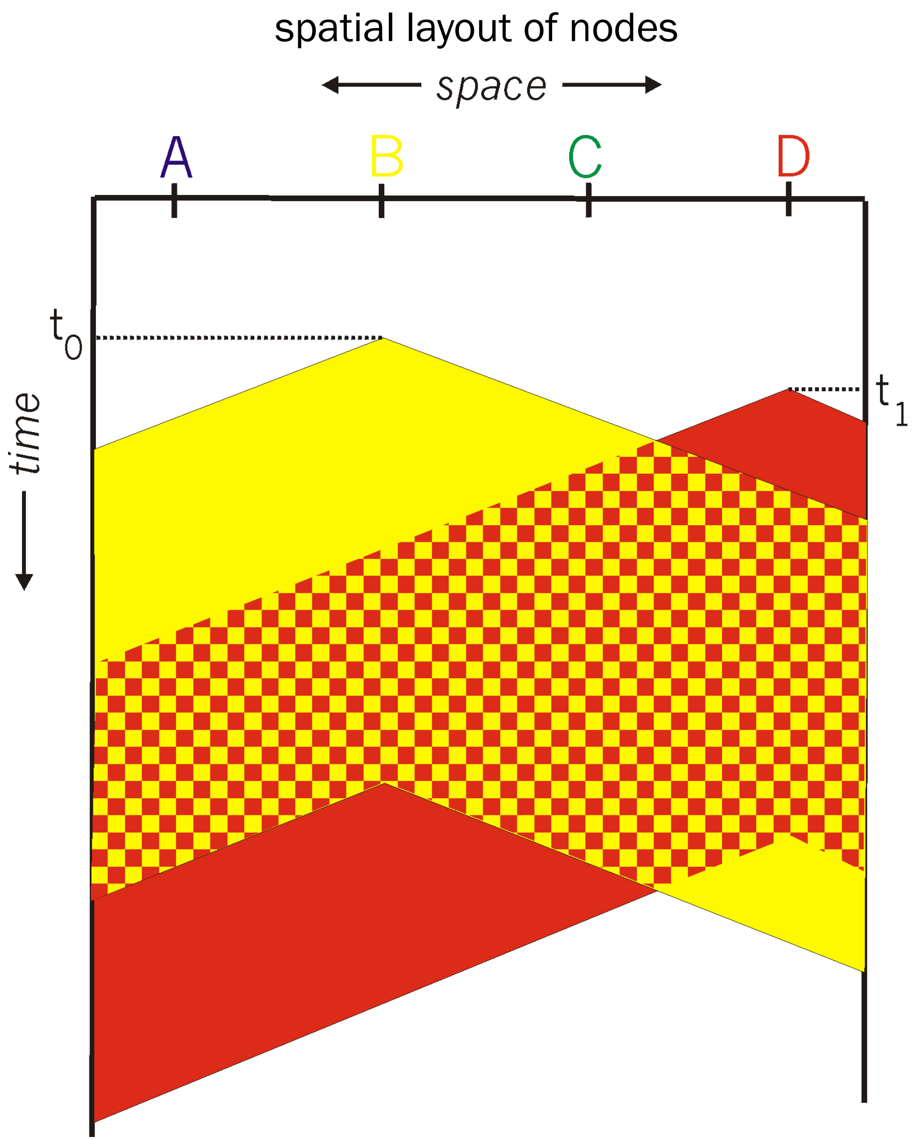
- CSMA: listen before transmit
- If channel sensed idle → transmit entire frame
- If channel sensed busy → defer transmission
- Human analogy: someone else talking? → Don’t interrupt!

CSMA COLLISIONS

Collisions can still occur:
Propagation delay means two nodes may not hear each other's transmission

Collision:
Entire packet transmission time wasted

Note:
Role of distance & propagation delay in determining collision probability



CSMA/CD (COLLISION DETECTION)

- CSMA/CD: Carrier Sensing as in CSMA
  - Collisions detected within short time
  - Colliding transmissions aborted, reducing channel wastage
- Collision Detection:
  - Easy in wired LANs
    - Measure signal strengths, compare transmitted, received signals
  - Difficult in wireless LANs
    - Received signal strength overwhelmed by local transmission strength
- Human analogy: the polite conversationalist
- CSMA/CD used in Ethernet 802.3 and CSMA/CA used in 802.11

REVIEW

SUMMARY

The student should be able to:

- Explain the services provided by Layers 1 and 2
- Describe the TCP/IP protocols commonly used at Network Access Layer specifically: ARP, Ethernet (IEEE802.3), WLAN and WiFi (IEEE802.11), Bluetooth (IEEE802.15) and WiMax (IEEE802.16)
- Simulate how ARP works in physical addressing and examine Ethernet frames
- Simulate how switches work to facilitate network connectivity using Packet Tracer